The Name-Leash Hypothesis:
Maternal Separation and the Origins of Displaced Reference

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This is version 3 of the preprint, expanding version 2 (Maslov, 2026, Zenodo, doi: 10.5281/zenodo.19145762). The argument is the same; this version adds three components: a finer analysis of displacement (the three levels of decoupling, §2.2), a tighter semiotic core (the broken-index repair, §3, including Wray's independent convergence), and a more developed comparative-neural anchor (§§8.3–8.4, 9.2, 9.4–9.5). A companion ***Supplementary Materials*** *document develops the intellectual genealogy of the displacement problem, the pre-cooperative-breeding ecology, the modal philosophy of the sign, and the cultural elaborations of the proto-name mechanism, all of which are referenced from the main text.*

Abstract

How did human communication first escape the "here and now"? Displaced reference — the ability to refer to entities outside the immediate perceptual field — is a defining feature of human language (Hockett, 1960), and Bickerton (2009) rightly made it central to the language-origins problem. Yet the usual recruitment-to-food scenario does not explain why a genuinely offline representational system would be selected when pointing or leading could guide others to a distant resource.

The Name-Leash hypothesis proposes a **proper name** — an individualized vocalization addressed to an absent infant — as a plausible candidate for the first displaced signal, rather than a food call. In cooperative-breeding contexts characteristic of early *Homo* (Hrdy, 2009; Burkart et al., 2025), mothers regularly transferred infants to allomothers and departed to forage. This created a recurrent affective pressure — separation anxiety directed at a specific, non-fungible child — that no perceptually grounded gesture could resolve.

The proposed semiotic mechanism is the **broken index**. When the referent of an indexical signal disappears, the index loses its perceptual anchor. A proto-name repairs this break by preserving the singularity of pointing while acquiring the context-independence of a symbol. As a rigid designator (Kripke, 1980), the name carries no descriptive content requiring environmental verification; it can therefore function precisely when there is nothing to perceive.

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The paper further distinguishes three levels of decoupling within displacement: arousal-decoupling, here-decoupling, and now-decoupling. The first two are documented in several animal systems, especially cooperatively breeding callitrichids; the third — maintaining an individual representation across a temporal gap — is the hominin-specific achievement the proto-name mechanism is designed to explain.

The hypothesis integrates evidence from cooperative breeding, upright walking and vocal-respiratory control, developmental phenomena such as solitary babbling and Fort/Da, comparative data on vocal individuation and mimesis, and neural evidence for cross-modal identity representation. It argues that the proto-name, born as affective self-regulation, became the minimal bridge from indexical signaling to displaced symbolic reference.

Keywords

Language evolution, displaced reference, cooperative breeding, proper names, concept cells, rigid designation, separation anxiety, vocal self-regulation, displacement activity, mimetic contagion, genitive construction, broken index, Default Mode Network, left temporal pole

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1. Introduction

One of the most persistent puzzles in language evolution is the origin of **displaced reference**: the ability to refer to entities not present in the immediate perceptual environment. The problem is functional and semiotic. In Peircean terms (Short, 2007), animal signals are predominantly **indexical**: they depend on a direct causal or perceptual relation to their objects and normally cease when the relevant stimulus disappears. A symbol, by contrast, can maintain its referential link independently of the object's perceptual availability. The transition from index to symbol therefore requires a mechanism by which a signal can survive the absence of what it refers to.

Existing models illuminate parts of this transition but leave a gap. Ecological recruitment (Bickerton, 2009), shared intentionality (Tomasello, 2008), the symbolic threshold (Deacon, 1997), motherese as audio leash (Falk, 2004, 2009), and holistic protolanguage (Wray, 1998, 2002) either presuppose some capacity to refer to absent entities or specify a logical structure without locating the concrete evolutionary pressure that made it chronic. The more elementary question is: what kind of act could first maintain a stable referent when the referent was no longer perceptually available?

The **Name-Leash hypothesis** answers with a candidate: a proper name — an individualized vocalization addressed to an absent infant. In cooperative-breeding contexts characteristic of early *Homo* (Hrdy, 2009; Burkart et al., 2025), mothers who transferred infants to allomothers and departed to forage faced the recurrent loss of perceptual contact with the infant. The hypothesis proposes that a vocalization originally embedded in infant-directed contact was rhythmically repeated in solitude as affective self-regulation. In doing so, it maintained the representation of a specific absent infant and became the minimal displaced referential unit.

The proper name is a strong candidate for this role because it designates a specific individual across contexts: it is a rigid designator (Kripke, 1980), rather than a description that must be verified against the environment. The neural machinery for cross-modal invariant identity representation was already available in the primate brain (Quian Quiroga et al., 2005; Tyree, Metke & Miller, 2023). What the hypothesis adds is a pathway connecting that machinery to cooperative breeding, maternal separation, vocal self-regulation, and eventual conventionalization in an alloparenting network.

The argument proceeds in twelve steps. Section 2 refines the concept of displacement by distinguishing escape from "here" and escape from "now," then introduces three levels of decoupling. Section 3 develops the broken-index mechanism and its convergence with Wray's account of proper names in holistic protolanguage. Sections 4–6 present the physiological, ecological, and semiotic core of the hypothesis. Sections 7–9 review developmental, comparative, and neural evidence. Section 10 treats the genitive contour as a proto-syntactic seed. Section 11 draws the anthropogenetic implications in compressed form. Section 12 compares the hypothesis with existing models. Extended philosophical, ecological, typological, and cultural discussions are developed in the Supplementary Materials.

Argument Map

The reader may find it useful to keep the following layout of claims in mind, distinguishing the core mechanism from its evidential supports and more speculative extensions.

- **Core claim.** A plausible candidate for the first displaced referential signal in human evolution was a proper name addressed by a mother to an absent infant under cooperative breeding (§§3, 5).
- **Mechanism.** The proto-name arose as rhythmic vocal self-regulation that repaired a broken index, conventionalized through mimetic contagion in the alloparenting network, and became a rigid designator without prior shared intentionality (§§4, 6).
- **Supporting evidence.** Developmental traces, comparative findings, and neural plausibility anchors jointly constrain — without proving — the proposed pathway (§§7–9).
- **Extensions.** From the proto-name follow a decomposition-first proto-syntax through the genitive contour (§10), and an integration with broader anthropogenesis in which reference, obligation, and time co-emerge (§11).
- **Supplementary expansions.** S1–S5 develop the intellectual genealogy, pre-cooperative-breeding ecology, modal philosophy of the sign, synchronic windows on the genitive contour, and cultural fossils of the mantra mechanism.

2. Displacement: The Hockett–Bickerton Framework Refined

2.1 Displacement as Escape from "Now," Not from "Here"

Hockett (1960) defined displacement as the ability to communicate about things "remote in space or time," and Bickerton (2009) made this feature central to the language-origins problem. But the standard formulation conflates two operations that, from an evolutionary standpoint, are radically distinct.

The escape from "here" was accomplished long before language. Any organism with self-locomotion and distance reception has already broken free from spatial immediacy: a hawk perceives a mouse at a hundred meters; a vervet monkey produces an alarm call in response to a leopard on a distant ridge. The referent is spatially remote — *not here* — but it remains within the perceptual field: it is *now*. In Korzybski's (1921) terms, animals are **space-binders**: to *bind* a dimension, in this sense, is to be freed from confinement within it, and animals — ranging freely through space — have escaped spatial immediacy. But they remain bound to time: everything within the perceptual field, however distant, is "now." Escape from "now" is what Korzybski reserved for the human **time-binder**, and it is this temporal release with which the present argument is concerned.

The escape from "now" is the distinctively human achievement. An entity that is not in the perceptual field in *any* modality — not visible, not audible, not detectable by smell — is not spatially remote; it is **temporally absent**. It exists for the vocalizer only as something that *was* (and may be again), or as something that *is now for someone else* (the child is with the allomother, perceptually inaccessible to me but presumed to exist in another's "now"). Displaced reference, properly understood, is the capacity to maintain the representation of an entity across a temporal gap — to bridge two "nows" that are not perceptually connected.

This distinction reframes the problem the Name-Leash hypothesis addresses. The "broken index" is not broken because the referent is far away — a loud cry might reach a distant infant. It is broken because the referent has *left the present*: the infant is beyond any perceptual horizon, in a "now" to which the mother has no sensory access. What must be bridged is not space but time. The maternal mantra is, in this precise sense, a **time-bridge** — a vocal act that sustains a representation *across* the present, linking the "now" of departure to the anticipated "now" of return.

2.2 Three Levels of Decoupling Within Displacement

The now/here distinction also clarifies why displacement should not be treated as a single threshold. A signal can be progressively detached from its original perceptual and motivational anchors along several independent axes, and each axis represents an evolutionary achievement in its own right. Three such axes are useful to distinguish:

Arousal-decoupling. The signal ceases to be a direct readout of the caller's internal arousal state. A vervet alarm call closely tracks the caller's level of fear; voluntary vocal control breaks this coupling, allowing a signal to be produced (or withheld) in disagreement with the underlying affective state. Without this decoupling, no further form of representational autonomy is possible: a signal that simply mirrors arousal cannot encode anything beyond it.

Here-decoupling. The signal can be produced without the referent being present in the caller's immediate perceptual field. This is the operation that distance reception accomplishes for many animal taxa, and it is integral to any system of contact calls used for "keeping in touch" (Falk, 2004). At this level the referent is no longer triggering the signal through perceptual presence; the signaling state, however, is still firmly in the caller's present — typically tied to a current contextual or motivational configuration.

Now-decoupling. The signal refers to an entity that is not in the caller's perceptual field in *any* modality and is not represented merely by the activation of a current categorical or motivational state. The referent must be held representationally — across a temporal gap, in a way that survives shifts in the caller's own context. The semiotic structure of this operation can be characterised, following West (2021), as an ego/non-ego dialectic — the signaler's current perceptual ground (ego) sustaining a held representation of the absent referent (non-ego) across the temporal gap. This is displacement in the strict, Hockett-Korzybski sense, and it is the level at which the Name-Leash hypothesis locates the proto-name. Now-decoupling presupposes the prior two levels but is not reducible to them.

These three levels are nested but distinct. Arousal-decoupling is a precondition for any communicative use of context that exceeds direct emotional broadcast. Here-decoupling is a precondition for signals that operate in the absence of their referents from the perceptual field. Now-decoupling is the additional, more demanding step at which a signal becomes capable of bridging two perceptually disconnected "nows." Section 8.3 examines the comparative evidence in light of this distinction, and argues that the first two levels are well documented in extant cooperatively breeding primates while the third is a hominin-specific achievement.

2.3 The Recruitment Model and Its Limitations

Bickerton's contribution is indispensable: he correctly insists that any serious account of language origins must explain how communication escaped the "here and now." However, the recruitment-to-food scenario does not, on its own, make this escape compelling.

First, the functional objection. Pointing and leading — both perceptually grounded — would seem sufficient for guiding conspecifics to a nearby carcass; it is not obvious why such a scenario would exert sustained selective pressure for a genuinely offline representational system. This objection receives empirical support from an unexpected quarter. Terrace (1985), analyzing the most advanced cases of symbolic communication in apes — Sherman and Austin's tool-exchange paradigm — showed that even in these cognitively sophisticated exchanges, all symbol use remained in the service of instrumental demands. If the richest symbolic behavior attainable by our closest relatives never escapes the demand register, the hypothesis that displaced reference arose from a more elaborate version of the same demand structure must explain why a qualitative threshold that apes never cross in the most favorable laboratory conditions would have been crossed by hominins facing the same functional pressure under less controlled environments.

Second, the semiotic objection. Bickerton rejected the indexical route as a dead end and proposed that displacement arose through **iconic** signs — sound imitations of absent objects. But this proposal contains a logical difficulty addressed in the next section.

Third, the recruitment model implicitly treats foraging as the primary evolutionary bottleneck. In early *Homo* there is no principled reason to privilege food acquisition over reproduction or offspring survival: any of these, if unmet, is fatal. The question is not which ecological problem was most important, but which recurrent situation most urgently demanded a signal that could remain active when its referent was gone.

The Name-Leash hypothesis proposes that this situation was not ecological but affective: the daily, non-negotiable physical separation between a mother and her non-fungible infant in a cooperative breeding context. The remainder of the paper develops the semiotic, physiological, ecological, and neural architecture of this proposal.

A fuller intellectual genealogy of the displacement problem is developed in Supplementary Materials S1.

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3. The Semiotic Argument: From Broken Index to Proper Name

3.1 Why the Iconic Path Fails

Bickerton, having dismissed the indexical route as a dead end, proposed that displacement arose through iconic signs — vocal imitations of absent entities. The idea is intuitive: a hominin imitates the sound of a distant animal in order to communicate about it in its absence.

This scenario contains a logical circularity. To imitate the sound of an absent object, the imitator must already hold a mental representation of that object in working memory — that is, the cognitive capacity for displacement must already be in place. The iconic sign does not *create* the ability to refer to the absent; it *presupposes* it. The icon is therefore a consequence of displacement, not its cause.

This leaves us with the index — the very route Bickerton rejected. But Bickerton's rejection was premature. He assumed that indexical signals are permanently locked into the "here and now" and therefore constitute a semiotic dead end. What he did not consider is the possibility that the index itself might be *transformed* under specific ecological and emotional pressures — not abandoned in favor of a new sign type, but reshaped from within.

3.2 The Broken Index

Consider the simplest communicative situation available to a social animal: an indexical gesture directed at a present object. A contact call, a reaching gesture, a gaze toward a companion — all are indexical in the Peircean sense: their meaning depends on a physical or causal connection to a present referent (Short, 2007).

Now consider what happens when the referent disappears. The mother turns to look for her infant, but the infant is no longer there — it has been taken by an allomother. The reaching gesture "hangs in the void." The contact call meets no response. The indexical sign has lost its object.

This moment of rupture — the **broken index** — is a semiotic crisis as well as a communicative failure. The communicative act demands completion; the affective system demands resolution. Something must fill the gap left by the absent referent.

3.3 The Proper Name as Repaired Index

The hypothesis proposes that this gap was filled by a vocal substitute: the rhythmic repetition of an individualized sound associated with the absent infant — a proto-name. This vocalization preserves the essential property of the index (it points to a specific, non-interchangeable individual) while acquiring a new property: it functions in the absence of that individual. It is, in effect, a **portable pointing gesture** — an index that has been detached from its perceptual anchor and converted into a symbol.

In Peircean terms, the proper name occupies a unique position in the typology of signs. It retains the singularity of the index (pointing to *this* individual, not a category) but operates through convention rather than physical contiguity (Short, 2007). It is, so to speak, the minimal symbol — the simplest sign that crosses the threshold from indexical to symbolic reference.

This semiotic analysis converges with Kripke's (1980) account of proper names as **rigid designators**: expressions whose reference remains fixed across all possible contexts. Crucially, a rigid designator carries no descriptive content that would need to be verified against the current environment: unlike "there is a leopard," which demands perceptual confirmation, a name activates its referent's representation independently of any state of affairs — making it the only signal type that can function precisely when there is nothing to perceive. This rigidity is the formal property needed to bridge the gap opened by the broken index.

A striking independent confirmation of this argument comes from evolutionary linguistics. Wray (1998, 2002), working within the holistic protolanguage framework, arrived at the same conclusion from a purely logical starting point: a pre-linguistic system consisting of unanalyzed holistic utterances (holophrases) breaks down precisely when a speaker needs to refer to an absent individual. Her analysis of the "displaced reference crisis" identifies the proper name — what she calls a "proper name through the back door" — as the only signal type capable of resolving this ambiguity. Wray's scenario posits that individualized commands (e.g., *baku* = "bring-Mary") were created for frequently needed calls to specific absent group members; over time, through reinterpretation in varying pragmatic contexts, the imperative shell eroded and *baku* came to mean simply "Mary" — a pure referential name (Wray, 2002, pp. 120–125). The convergence is significant: from the logic of holistic communication (Wray) and from the biology of maternal attachment (the present hypothesis), the same semiotic object — the proper name of a specific individual — emerges as the minimal unit capable of crossing the displacement threshold. The two accounts differ in what they supply. Wray identifies the *logical structure* of the transition but leaves open the question of *who* creates this first name, *for whom*, and under what affective pressure. The present hypothesis fills these slots: the mother creates the name for the absent infant, driven by separation anxiety within the cooperative breeding network. Wray describes the grammar of the transition; the Name-Leash hypothesis describes its biology.

3.4 Why Not a Common Noun?

On this hypothesis, the first displaced signal is more plausibly a proper name than a categorical label, for a convergent set of reasons. A common noun classifies — it applies to any member of a category, requiring prior abstraction over multiple instances; a proper name requires no such abstraction, since it refers to a single known individual already represented by a dedicated neural ensemble (a concept cell, §9.1). The name attaches a vocal label to an identity already encoded; it does not create a new representation. The Peircean semiotic substrate of this individuation-without-classification character is developed by West (2014), whose analysis of Peirce's "matrix of individuation" shows proper names and pronouns sharing an attentional anchor on the singular referent that common nouns lack. As a rigid designator, it carries no information about the environment and therefore requires no perceptual check — it can operate in the complete absence of perceptual access to its referent. Unlike a common noun, the proto-name requires no lexical category, no argument structure, and no selectional restrictions — what Tallerman (2014) calls *edge features*; in Jackendoff's (2002) terms it is a "defective lexical item" that has phonological form and a referent but no syntax, distinguished from other defective items (*ouch*, *hello*, *shh*) by being referential. Edge features accrue later, as the proto-name enters combinatorial contexts through externalisation in the alloparenting network (§10.3).

The motivational picture supports the same conclusion. Separation anxiety and attachment provide the affective subsidy that sustains the representation of an absent referent: what would otherwise be an effortful cognitive act is driven automatically by the limbic system, and the cost of *not* maintaining the representation (unregulated anxiety) exceeds the cost of maintaining it. Empirically, Terrace's (1985) review of ape-language research shows that the highest level of symbol use achieved by non-human primates — including categorisation of lexigrams — operates exclusively at the level of common-noun classification; apes learn to label categories but do not spontaneously name individuals. Only the non-fungible referent — the specific infant, irreplaceable and emotionally charged — generates sufficient endogenous pressure to sustain a representation without external reward. Finally, the proper name is the ideal *transitional* signal because it asserts nothing about the environment: it carries referential force without propositional liability, and there is therefore nothing to falsify and no arms race between deceiver and detector. Wray (2002) arrives at a complementary observation: manipulative holophrases lack truth values altogether. Truth and falsehood become possible only later, when decomposition into subject and predicate creates the propositional format.

3.5 Arbitrariness as Consequence: Saussure Inverted

The same broken-index mechanism also offers a genetic reading of Saussurean arbitrariness. As long as a sign remains indexical — a gaze, a reaching gesture, a contact call tied to a present referent — its form is motivated by the causal texture of the encounter. When the referent disappears and the vocalization persists, the signal both becomes displaced and loses its direct causal motivation. Displacement and arbitrariness are thus not two unrelated features of language; they are two consequences of the same rupture viewed from different angles.

This reading helps explain why proper names have often appeared anomalous within synchronic grammar. They behave like "defective" expressions only if common nouns are treated as the norm. From the present perspective, the proper name is not a defective common noun but a pre-grammatical sign: a minimal symbol that preserves the singularity of an index while becoming portable across contexts — what Sinha (2004) terms a *protosymbol* (conventional and arbitrary, displaying joint reference, but not yet integrated into a system of internal logical relations among signs). The later lexicon generalizes the arbitrariness first visible in personal names. A fuller cultural-semiotic discussion is given in Supplementary Materials S1.8.

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4. Physiological Prerequisites: Upright Walking as Exaptation for Vocal Self-Regulation

The semiotic argument identifies the *logical* structure of the transition from index to symbol. But logic alone does not produce vocalizations. The broken index could only be "repaired" by a proto-name if the hominin body was capable of sustaining rhythmic, voluntary vocal output independently of locomotion. This capacity was provided by the suite of physiological changes associated with habitual upright walking.

Decoupling respiration from locomotion. In quadrupedal mammals, locomotion and breathing are mechanically coupled: each stride compresses the thorax and forces an exhalation. Habitual upright walking broke this constraint. By freeing the thorax from the mechanical rhythm of the forelimbs, bipedalism permitted independent, voluntary control of expiratory timing and duration (Provine, 2000). This was the first prerequisite for any form of sustained rhythmic vocalization.

Orthostatic stress and autonomic adaptation. Upright posture imposed a novel cardiovascular challenge: maintaining cerebral perfusion against gravity. The baroreflex and autonomic adaptations required to solve this problem — enhanced vagal tone, refined heart-rate variability regulation — created an autonomic infrastructure that could be co-opted for other forms of stress management. In particular, the ventral vagal complex, which in Porges's (2011) polyvagal framework integrates laryngeal and facial motor control with parasympathetic regulation, became available as a platform for vocal affect regulation. Empirical evidence links prolonged, paced exhalation and rhythmic vocalization to increased parasympathetic influence and modulations in heart-rate variability (Bernardi et al., 2001; Vickhoff et al., 2013; Gerritsen & Band, 2018). These findings should be interpreted as physiological correlates rather than validations of any single theoretical framework; nonetheless, the convergence across studies of mantra recitation, choral singing, and controlled breathing strongly suggests that rhythmic vocal-respiratory patterns can modulate autonomic state.

The platform for vocal self-regulation. The result was an unprecedented capacity for self-soothing through vocalization. A mother could now regulate her affect while foraging at a distance from her infant; an infant, left temporarily with an allomother, could self-regulate through babbling. The hominin substrate for this mechanism includes voluntary control of expiratory timing, laryngeal motor pathways providing strong proprioceptive feedback to brainstem autonomic centers, and baroreflex adaptations that permit sustained activity while maintaining autonomic stability. This suite of changes — originally selected for the biomechanical demands of upright walking — constituted an **exaptive platform** for vocal self-regulation. Bipedalism did not evolve "for" language; but without the respiratory and autonomic freedoms it provided, the name-leash mechanism could not have been enacted.

The paleoanthropological record is consistent with this timing. By approximately 2.0–1.8 million years ago, *H. ergaster/erectus* exhibits full commitment to terrestrial bipedalism, orthostatic cardiovascular adaptations, and the earliest archaeological evidence for "home base" behavior (Oldowan sites). This is the same window in which cooperative breeding would have generated the selective pressure for daily, repeated separations between foraging mothers and cached infants.

A note on prior ecological context: the obstetrical-dilemma framing of bipedality (the constraint a narrow pelvis places on fetal brain size) treats upright walking primarily as a constraint on gestation. Jordania (2017) has argued, on independent grounds, that several distinctively human traits — habitual bipedalism, loud rhythmic vocalization, stone-tool use, body painting, and group synchrony — originated as components of an integrated **aposematic defense strategy**, a collective display designed to deter predators on the open savanna. This framing has the advantage of providing an ecological account of the very autonomic adaptations on which the present argument depends: the upright stance is simultaneously a threat posture and a stress posture, and the resulting selection pressure for vagal regulation is what later supports vocal self-regulation. The aposematic frame also generates a plausible pre-CB infrastructure (synchronic male coalitions, scavenging niche, costless commensality) from which cooperative breeding could subsequently emerge. The fuller development of this argument is given in Supplementary Materials S2; here the point is simply that the physiological exaptation described in this section may belong to the pre-linguistic ecology itself rather than to a fortunate coincidence.

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5. Cooperative Breeding and the Affective Gap

5.1 The Cooperative Breeding Model

The preceding sections have identified the semiotic structure (broken index → proper name) and the physiological platform (upright walking → vocal self-regulation) required for displaced reference. What remains is the ecological context that would generate sufficient and sustained evolutionary pressure for such a mechanism to be selected. The Name-Leash hypothesis locates this pressure in cooperative breeding.

Hrdy (2009) argued that early *Homo* was characterized by **cooperative breeding** (alloparenting) — a reproductive strategy in which individuals other than the biological mother contribute substantially to infant care. Unlike the mother-centric rearing typical of other great apes, cooperative breeding involves the regular transfer of infants to allomothers (older siblings, grandmothers, other coalition members), freeing the mother for extended foraging. Recent work by Burkart, Cerrito, Natalucci, and van Schaik (2025) strengthens this model by treating cooperative breeding as a likely **early catalyst** of the human evolutionary feedback loop, rather than as a later consequence of increasing brain size and altriciality. If this chronology is correct, cooperative breeding was in place by approximately 2 million years ago — contemporaneous with the bipedal and home-base adaptations of §4.

A terminological clarification. Among primates, only callitrichids qualify as cooperative breeders in the strictest physiologically-obligate sense (Burkart, van Schaik & Griesser, 2017). Humans practice a form of allomaternal care that is pervasive and evolutionarily consequential but not strictly obligate at the individual level; specialists increasingly use softer terms for the human case — "cooperative childcare," "facultative" cooperative breeding, or "communal breeding" (cf. Kramer, 2010; Hrdy, 2009). Unless otherwise specified, "cooperative breeding" in what follows refers to the human variant of extensive but facultative allomaternal care. The distinction matters: the forms of affective pressure generated by facultative human CB differ systematically from those generated by obligate callitrichid CB, and the Name-Leash mechanism is specifically adapted to the former — a point developed in §5.2 and §8.3.

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The cooperative breeding hypothesis as applied to human evolution is not a single theoretical position but a spectrum of models, ranging from the classical Bowlbyan dyadic frame and the pair-bond / paternal-provisioning model (Lovejoy, 1981) to the grandmother hypothesis (Hawkes, 1997; Hawkes et al., 1998) and the networked-CB position (Hrdy, 2009; Burkart et al., 2017, 2025). The Name-Leash hypothesis adopts the networked position and depends critically on the *network* character of CB rather than on its precise demographic intensity: the conventionalization step the hypothesis describes (§6.5) cannot operate in a fixed dyad, since it is the diversity of pragmatic perspectives that drives the segmentation of the holophrastic proto-name. A fixed pair-bond regime, or a regime in which alloparental input is occasional and limited to one or two figures, provides too few perspectives. A fuller positioning within this spectrum is given in Supplementary Materials S2.

5.2 The Affective Gap: Separation from a Non-Fungible Infant

Cooperative breeding generates a specific cognitive-affective configuration unique among primates. The mother regularly transfers her infant to an allomother and physically separates from it, but the **affective bond** with the specific infant — driven by attachment, oxytocin, and selection pressure on parental investment — is not extinguished by separation. The infant is **non-fungible**: it cannot be exchanged for another, even a related one. This non-fungibility is the cognitive translation of the biological fact that natural selection operates on the survival of one's own offspring.

The result is a daily, recurrent affective tension: a strong emotional bond directed at a specific individual physically absent for hours at a time. This tension has no analogue in other great apes, where mothers do not relinquish their infants. It is precisely the kind of recurrent, high-investment, high-cost situation that exerts sustained selective pressure for a specialized cognitive solution.

A clarification is in order. The Name-Leash hypothesis does not predict that contemporary CB-adapted human mothers will exhibit acute reactive distress at every routine separation; the very mechanism the hypothesis describes is what *makes such separations tolerable*. The argument is structural rather than reactive. Selection acts on the architectural problem of *holding a specific, non-fungible referent across an interval of perceptual suspension* — the problem captured by the formula *my child is not with me*. The phrase is used here in a relational sense: the child has not disappeared and is not simply alone somewhere, but is outside the mother's immediate keeping and held within another caregiver's present. This is a **structural pressure** even where it is not a reactive one. The proto-name renders separation tolerable by holding the referent's representation stable across an absence that the great-ape attachment baseline would otherwise make difficult to sustain. The hominin lineage, on this reading, entered cooperative breeding from a strong-attachment baseline that it retained rather than dismantled, and the proto-name mechanism is the semiotic accommodation that made the new arrangement viable. Supplementary Materials S2 develops the contrast between this **facilitating** route (hominins) and the **forcing** route to cooperative breeding (callitrichids, where obligate twinning makes solitary rearing physiologically impossible) and what each implies about separation tolerance.

A second clarification concerns distribution. Burkart and colleagues observe that in CB species, vigilance and affective investment in the infant are not concentrated in the mother but *distributed* across the group. This distributed pressure gives the hypothesis exactly the network conditions required for mimetic spread (§6.5). The proto-name emerges not because the mother alone is in crisis, but because the mother–infant separation constitutes a problem-state for the whole alloparenting network, and the vocal solution, once adopted by one member, is subject to contagion across the group. The distribution of care-motivation, documented hormonally by Finkenwirth et al. (2016), is the substrate on which mimetic conventionalization operates.

5.3 The Affective Gap vs. The Ecological Gap

The difference between the Name-Leash hypothesis and Bickerton's recruitment model can be stated precisely in terms of the referent's properties:

Feature	Bickerton's Recruitment Model	Name-Leash Hypothesis
Referent	Food source / Predator ("it")	Specific infant ("you")
Fungibility	Replaceable	Non-fungible
Frequency	Episodic, opportunistic	Daily, recurrent
Emotional valence	Moderate (hunger, fear)	Maximal (attachment, separation anxiety)
Duration of need	Minutes (until carcass exploited)	Hours (entire foraging bout)
Information content	High (location, type, danger)	Minimal (identity only)
Deception structure	Adversarial (false claim, conflict of interest)	Self-regulation (no conflict; implicit promise of return)
Decoupling axis (§2.2)	Here-decoupling (referent at distance)	Now-decoupling (referent across temporal interval)

The hypothesis identifies a referent that is non-fungible, recurrently absent, maximally emotionally charged, and informationally minimal — precisely the configuration that exerts selective pressure for a context-independent rigid designator rather than for a perceptually grounded recruitment signal.

5.4 Trust, Transfer, and the Proto-Credit Operation

The transfer of an infant to an allomother is itself a remarkable social act. The mother entrusts her non-fungible child to another individual, with the implicit expectation that the child will be returned. This structure resembles a **proto-credit operation**: the mother "lends" the child, the allomother "holds" it in trust, and the obligation of return is implicit in the relationship.

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This observation will become important when we consider the genitive contour (§10) and the broader anthropogenetic implications (§11). For now the key point is that cooperative breeding generates both separation anxiety and a relational structure (A entrusts X to B; B holds X for A; B returns X to A) that is already, in embryonic form, the structure of social obligation.

Independent ethnographic evidence supports the structural claim that the infant transferred to alloparents is held within the network as an *inalienable possession* — circulated but not detached from its source. In Samoan kinship usage, the child given by a mother to her brother for fostering is named *tonga*, the same term used for the inalienable possessions that, in Mauss's classic analysis (1925/1990), retain their compelling-return force across cycles of giving and receiving (Weiner, 1992). The terminology is explicit: the alloparented child is transferred but not given, present in another's care but bound to the maternal source. This is precisely the structural profile the proto-credit operation requires.

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6. The Core Mechanism: From Affective Mantra to Displaced Symbol

The argument of this section can be stated as a four-step minimal mechanism. The remainder of the section unpacks each step in turn.

Step 1 — Self-directed loop. A foraging mother, separated from her non-fungible infant, rhythmically repeats a vocalisation associated with that specific child. The pattern is initially self-regulatory: it modulates her autonomic state through paced respiration (§4) and sustains the infant's neural representation through identity-selective concept-cell activity (§9.1) in the absence of any perceptual cue.

Step 2 — Shared rhythm. The vocalisation is not silent. Heard by allomothers under conditions of elevated arousal — themselves engaged with the same infant — it acts as an attention-capturing displacement activity (§6.4) and is reproduced through phylogenetically older mimetic mechanisms. Reproduction is reinforced immediately by the infant's calming response.

Step 3 — Form stabilisation. Across a network of multiple alloparents interacting with the same infant on different occasions, the vocal pattern is exposed to the perspectival diversity that drives mimetic conventionalisation (§6.5). The form is sharpened to the invariant that all caregivers' uses share; the situational variability is filtered out.

Step 4 — Referential fixation. What remains, at the convergence point of multiple perspectives on the same infant, is a vocal form whose use is no longer contingent on any one caregiver's emotional state and whose reference is no longer contingent on the infant's perceptual presence. The form has become a rigid designator (Kripke, 1980) — the proto-name.

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The mechanism does not presuppose shared intentionality, propositional content, or grammatical apparatus. It requires a sufficiently imitogenic signal, a reinforcement mechanism (the infant's differential response), and a sufficiently dense alloparenting network. Each is supplied by the cooperative-breeding ecology developed in §5.

6.1 The Mantra Function

A mother in a cooperative breeding coalition leaves her non-fungible infant with an allomother and departs to forage. The infant is out of sight, out of earshot. The indexical connection — contact calls, gaze, touch — is broken. The mother faces a prolonged severance of perceptual contact with her infant.

To mitigate the resulting autonomic stress, the mother begins to rhythmically repeat a vocalization associated with her child — an individualized sound originally used in face-to-face infant-directed interaction. This is the proto-name: a motherese-modified vocal signature of the specific infant.

The repetition performs a dual function. **Physiologically**, it constitutes a rhythmic vocal-respiratory pattern that modulates autonomic state — the self-soothing mechanism described in §4. Paced exhalation and laryngeal proprioceptive feedback increase parasympathetic tone, reducing the physiological correlates of anxiety. **Cognitively**, it activates and sustains identity-selective neurons (concept cells) in the medial temporal lobe (Quian Quiroga et al., 2005; Quian Quiroga, 2012). By rhythmically activating this neural ensemble in the absence of any perceptual stimulus, the mother creates a **neurocognitive leash** — a sustained mental representation of the absent child.

This is the name-leash: a vocal act that keeps the absent referent "present" through the combined operation of autonomic self-regulation and identity-specific neural activation.

The neurocognitive status of this proto-name can be specified through the dual-process model of language (Wray, 2002, 2008; Van Lancker Sidtis, 2004, 2021), according to which the human brain processes language through two parallel systems: an analytic system (left-hemispheric, generating novel utterances through grammatical rules) and a formulaic system (right-hemispheric and subcortical, storing and retrieving holistic, overlearned vocal units — greetings, oaths, expletives, prayers, and proper names — as unanalyzed chunks). The proto-name is, by this classification, a **formulaic expression** *in statu nascendi*. It is holistically produced, affectively charged, automatically retrieved through limbic activation, and stored as a unitary motor program in the basal ganglia. This formulaic character explains why the proto-name could be stable and transmissible *before* the emergence of analytic grammar (it did not depend on grammar, being processed by a phylogenetically older system); why proper names are preserved in cases of severe left-hemisphere aphasia while common nouns are lost (Van Lancker & Klein, 1990); and why the proto-name could function as a self-regulatory mantra (its production engages the same subcortical–limbic circuitry that mediates emotional vocalization in all mammals).

A computational consideration supplements the affective-physiological rationale. Research on visual and mental attention (Pylyshyn, 1989, 2000; Luck & Vogel, 1997; Cowan, 2001) converges on an approximate capacity limit of **four pre-attentive deictic indexes** — Pylyshyn's "FINSTs" — and Hurford (2003) invokes this limit to explain why natural-language clauses rarely exceed three or four principal participants. A foraging mother must track many referents at once — terrain, food, group members — so these few slots are quickly saturated; under active foraging they are preferentially allocated to the current sensorimotor task, and the absent infant is precisely the referent that cannot be maintained through ordinary working-memory resources. The mantra solves this by moving the maintenance of the absent referent *outside* the four-FINST workspace: a rhythmic vocal-respiratory pattern operates through limbic and brainstem machinery rather than consuming an attentional index. The vocal character of the proto-name is, on this reading, not incidental but structural: silence would not suffice. Modern human working memory may, on this account, be a hybrid architecture in which the four-slot pre-attentive index system is supplemented by a phonologically mediated "name buffer" that runs on the neural infrastructure originally recruited by the maternal mantra — Baddeley's (2012) phonological loop as the domesticated descendant of the name-leash.

6.2 The Birth of Displacement

This moment is the precise point at which displacement occurs. The vocalization no longer reacts to a present stimulus; it **generates a mental state to compensate for a missing one**. The sound is not triggered by the environment — it is produced *against* the environment, to counteract the absence that the environment imposes. In semiotic terms, the broken index has been repaired: the vocal gesture that once pointed at a present infant now sustains the representation of an absent one. In Kripke's (1980) terms, the proto-name has become a rigid designator: its reference is fixed to a specific individual independently of any contextual state of affairs. The name does not describe where the child is; it simply **holds the identity**.

6.3 The Symbolon

The ancient Greek *symbolon* — a token broken in two, each half held by a different party as a sign of mutual recognition — provides a suggestive analogy. The mother's vocalization is one half of the symbolon; the infant (held by the allomother) is the other. Neither half is meaningful in isolation; when the halves are brought together — when the mother returns and the name meets its bearer — the circuit of recognition is completed.

This analogy captures the essential relational structure of the proto-name: the name exists *for* reunion. It is not a label affixed to an object for purposes of classification; it is one half of a broken bond, maintained in anticipation of the bond's restoration. The name is, from its origin, a **promissory sign** — a token whose very existence presupposes the future act of reunion.

6.4 The Mantra as Displacement Activity

Ethologically, the maternal mantra may be understood as a displacement activity in the classical ethological sense (Tinbergen, 1952; Delius, 1967): a behavior released when two incompatible action systems are simultaneously activated. The mother must forage, but the attachment system pulls her toward the absent infant. Direct care is unavailable; foraging cannot be abandoned without cost. Rhythmic vocalization supplies a third channel through which the unresolved attachment tension can be discharged and regulated.

This framing is useful because it avoids treating the proto-name as a communicative invention from the start. Its first function is affective self-regulation; its referential function emerges because the vocal act is organized around one specific absent individual. The same structure has psychoanalytic and cross-modal analogues — transitional objects, repetitive play, rocking, bead-counting, and other rhythmic substitutions — but these are better treated as later or parallel elaborations rather than as separate arguments in the main text. The cultural and material extensions of this mechanism are developed in Supplementary Materials S5.

6.5 Mimetic Contagion in the Alloparenting Network

The maternal mantra is a paradigmatic displacement activity, and therefore a paradigmatic target for imitative contagion. Its conventionalization proceeds not through any cognitive agreement or shared understanding, but through the blind mechanism of mimesis operating within the dense social network created by cooperative breeding.

The **network** character of this process is essential. Cooperative breeding in early *Homo*, as modeled by Hrdy (2009) and Burkart et al. (2025), was not a fixed triad but a dynamic, context-shifting network of alloparents — what Hrdy describes as a “market” of care, in which multiple group members of both sexes and various ages participate in infant-directed care at different times.

A clarification about the kind of mimesis at issue. The contagion described here is the involuntary reproduction of an attention-capturing vocal pattern under conditions of elevated arousal. It does not require — and the argument does not depend on — the more demanding capacities sometimes also gathered under the term *imitation*: imitation of a planned action sequence, true imitation of intent, or the kind of culturally cumulative imitation that Donald (1991) places at the centre of his “mimetic culture” stage. The kind of mimesis the proto-name’s conventionalization rides on is the simpler, phylogenetically older capacity for echopraxic and echolalic reproduction of conspicuous patterns under emotional load, of the kind extensively documented in primate species including marmosets and tamarins (Snowdon & Elowson, 2001; Voelkl & Huber, 2007).

The mechanism proceeds in four steps.

(a) Primary contagion. The allomother, holding a distressed infant, is herself in a state of elevated arousal — the universal mammalian response to infant distress (Hrdy, 2009).

Elevated arousal suppresses prefrontal inhibitory control, releasing the phylogenetically older imitative mechanisms mediated by the mirror neuron system (Rizzolatti & Craighero, 2004). When the mother's mantra is heard, the allomother reproduces the vocal pattern — not because she "understands" its referential function, but because the sound is a displacement activity heard during a state of high arousal. The allomother's reproduction produces an immediate reward: the infant, recognizing the familiar vocal pattern, calms. This creates a positive reinforcement loop that stabilizes the behavior.

(b) Infant-mediated selection. The infant functions as an unwitting arbiter of vocal fidelity. Accurate reproduction of the mother's vocal signature produces calming; inaccurate reproduction does not. This creates selective pressure for precise inter-individual vocal matching — a pressure that operates automatically through the infant's differential response, without any party needing to evaluate or enforce accuracy.

(c) Network amplification. In a dynamic alloparenting network, the same infant may be held by three, five, or more different caregivers over a period of days. Each caregiver independently acquires the proto-name through the same mimetic-reinforcement mechanism. The result is that multiple individuals now produce the same individualized vocalization when interacting with the same infant — a *de facto* convention that has arisen without any convention-forming act.

(d) Cross-exposure and paradigmatic contrast. A single allomother who cares for more than one infant over time is exposed to *multiple* proto-names, each associated with a different child. This cross-exposure is the precondition for the emergence of paradigmatic relations and ultimately for the decomposition of the holophrastic proto-name into combinatorial elements (§10).

6.5.1 The Intergenerational Bottleneck

The alloparenting route also solves a transmission problem. A signal invented only in adult foraging would have to be learned later by infants without being embedded in their primary learning ecology. The name-leash, by contrast, originates in the caregiving network itself: mothers, allomothers, siblings, and infants all participate in the same vocal loop. The innovation is therefore available at both critical stages — adult use and infant acquisition — which makes conventionalization more plausible than in scenarios where the signal belongs only to adult coordination.

6.6 Double Attraction: The Infant as Irresistible Focus

The mechanism above is powered by a double attractor structure. The proto-name, as a displacement activity, possesses heightened imitogenic potential (Porshnev, 1974). The infant itself, under conditions of cooperative breeding and the neotenization characteristic of the hominin lineage, constitutes an object of irresistible attraction for all members of the group: the Kindchenschema response (Lorenz, 1943) is intensified in humans by neotenization (Hare, 2017), itself underwritten by transcriptional retention of juvenile gene-expression patterns in the human prefrontal cortex (Somel et al., 2009), and under cooperative breeding the infant becomes the gravitational center of social attention. It is instructive to compare this attractor with two others. An attractive mate is a powerful focus of attention but generates competition: access is zero-sum. The infant generates cooperative rather than competitive attention; each additional caregiver is a resource, not a rival. The infant combines the properties needed for the birth of a conventional signal: **individuality** (unlike fire, which is generic), **universality of appeal** (unlike the mate, whose attraction is segmented), and **non-rival access** (unlike the mate, whose access is zero-sum).

6.7 Convention Without Understanding

A potential objection: if conventionalization requires "shared intentionality" in Tomasello's (2008) sense — two or more agents jointly attending to the same referent with mutual awareness of each other's attentional states — then the mechanism described here would presuppose the very cognitive capacity it is supposed to explain. The response is that conventionalization, as described here, **does not require shared intentionality as a precondition**. It requires only three things: a sufficiently imitogenic signal, a reinforcement mechanism (the infant's differential response), and a sufficiently dense social network.

What Tomasello describes as "shared intentionality" can, on this framework, be understood as an **emergent property** of the process rather than its precondition. When multiple alloparents independently acquire the same proto-name through mimetic contagion, and each uses it in interaction with the same infant, a *de facto* convergence of perspectives on a single object has been achieved. The infant has become the **invariant** at the intersection of multiple viewpoints — each caregiver's distinct perspective converging on the same individual through the medium of the shared sound. Object invariance — the sense that "this is the same thing" across different perspectives — is not a cognitive prerequisite for this convergence; it is its *product*. The Name-Leash mechanism therefore changes the explanatory order: symbolic convergence need not presuppose shared intentionality; it may help produce it.

6.8 Why Vocalization Prevails

While the core mechanism — rhythmic, self-directed behavioral iteration that maintains the absent referent in working memory — is formally neutral with respect to modality, vocalization is the most parsimonious candidate. Vocal self-regulation leaves no archaeological trace, requires no tools, permits simultaneous foraging, and exploits the respiratory-autonomic coupling created by upright walking. Alternative modalities may have played supplementary roles, but the vocal route combines physiological efficiency, social transmissibility, and compatibility with mobile foraging in a way no other modality matches.

Figure 1 summarizes the proposed causal chain.

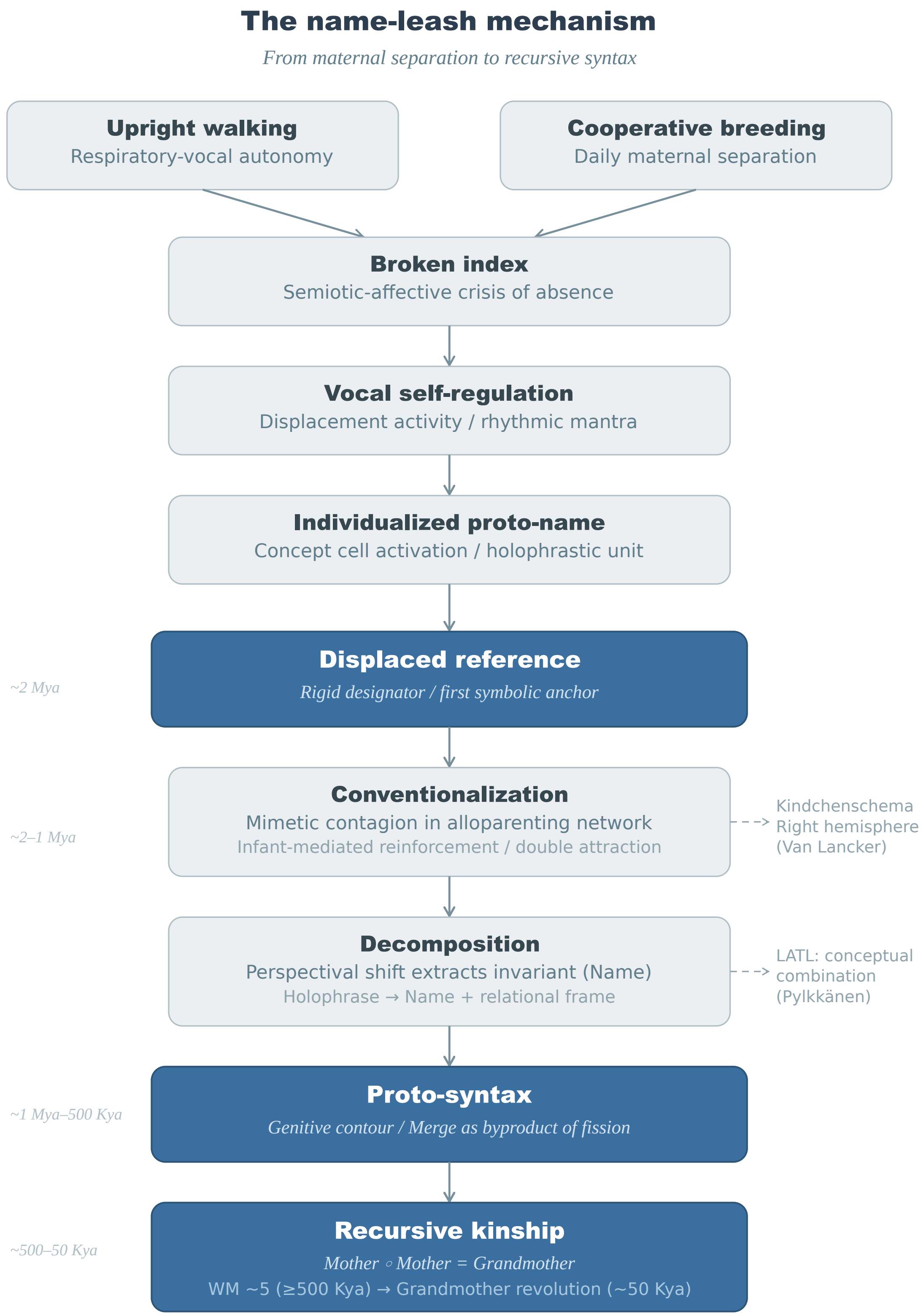

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Figure 1. *The Name-Leash mechanism.* Two independent evolutionary prerequisites — respiratory-vocal autonomy from upright walking and daily maternal separation from cooperative breeding — converge to produce a semiotic crisis: the broken index. Rhythmic vocal self-regulation repairs this break by sustaining the absent referent's neural representation via concept cell activation, producing the first displaced reference — a rigid designator functioning as the initial symbolic anchor. Conventionalization through mimetic contagion transforms the private mantra into a shared symbol, whose inherent relational (genitive) structure provides the seed for proto-syntactic Merge.

7. Developmental Evidence: Babbling, Fort/Da, and the First Word

The Name-Leash hypothesis proposes a phylogenetic scenario. If the scenario captures a genuine cognitive-affective dynamic, traces should be visible in ontogeny — in the developmental sequence by which human infants acquire their earliest symbolic capacities. Three independent bodies of developmental evidence converge on the same pattern.

7.1 The Babbling Peak and Separation Anxiety

Delack's (1976) longitudinal data on infant vocalizations, as compiled in Locke (1993), document a striking pattern across the first year of life (Figure 2). Between 33 and 43 weeks — the peak window for separation anxiety (Bowlby, 1969; Schaffer & Emerson, 1964) and the emergence of representational object permanence (Piaget, 1954) — **solitary vocalizations** peak at approximately 67% at week 38, while vocalizations in the mother's presence reach a sustained minimum of approximately 20% across weeks 33–43.

This temporal alignment is unlikely to be coincidental. At precisely the developmental stage when the infant becomes capable of holding an absent caregiver in mind and simultaneously experiences the greatest distress at the caregiver's absence, the infant dramatically increases its rate of self-directed vocalization. The pattern suggests a functional role for babbling in **infant self-regulation**: the vocal-respiratory cycles of canonical babbling may serve as articulatory practice and as an emergent self-soothing mechanism that bridges the affective gap of the caregiver's absence (Stark, 1986).

7.2 Freud's Fort/Da

In *Beyond the Pleasure Principle* (1920), Freud describes a game observed in an eighteen-month-old child. The child repeatedly throws a wooden spool attached to a string, uttering "Fort!" ("gone!"), and then pulls it back, exclaiming "Da!" ("there!"). Freud interprets this as the child's symbolic mastery of the mother's departure and return. For our purposes, the critical observation is structural: the minimal symbolic act arises from the experience of absence, and its function is not referential but **regulatory** — it allows the child to manage the affect generated by separation. The game's structure — presence → absence → return — is the minimal unit of diachrony.

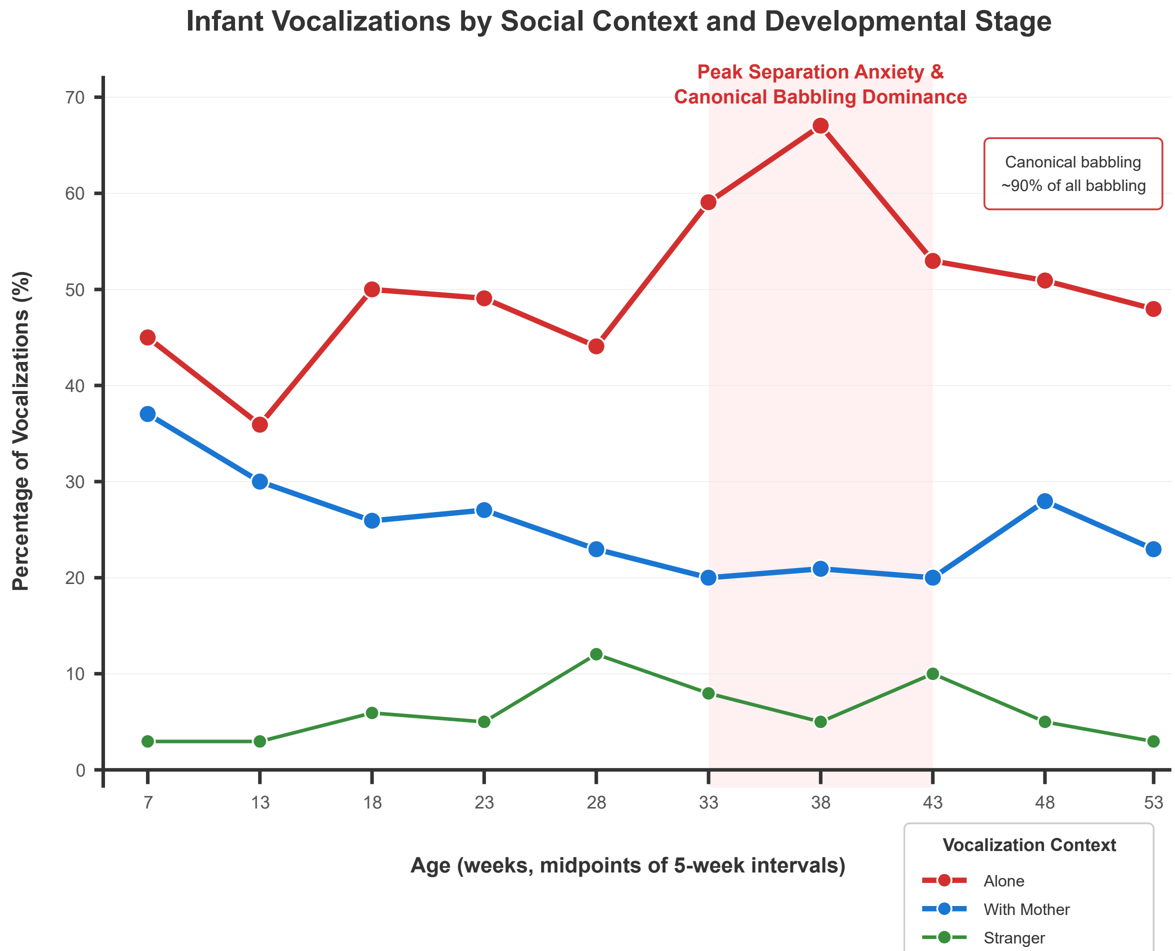

Figure 2. *Infant vocalizations by social context across the first year of life.* Data show the percentage of infant vocalizations occurring when alone (red), with mother (blue), or with a stranger (green). Data from Delack (1976), as compiled in Locke (1993, Fig. 4.1). The shaded zone (33–43 weeks) marks the critical window of separation anxiety and canonical babbling dominance, during which solitary vocalizations peak (67% at week 38) and maternal-presence vocalizations reach their minimum (~20%).

7.3 Porshnev's "First Word"

A complementary observation comes from Porshnev (1974), who notes three distinctive features of the child's first articulated word, typically appearing at 11–13 months: **the first word is singular** (a lone vocal act, separated from the second word by a considerable pause); **the first word is functionally regulatory, not referential** (whatever the phonetic content — "mama," "clock," "kitty" — the function is self-inhibition through vocalization, the imitation of a sound complex associated with the inhibition of the child's own reaching, grasping, or manipulative actions); and **the prototypical situation is one of separation from the mother** — the most potent prohibitions in this developmental window involve

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the denial of maternal contact (weaning, refusal to hold, departure). This is the ontogenetic analogue of the separation stress that the Name-Leash hypothesis posits as the phylogenetic driver.

7.4 Convergence

Three independent lines of developmental evidence converge on a single pattern: the earliest symbolic or proto-symbolic vocal acts arise in the context of **maternal separation**; their primary function is **affective self-regulation**, not information transfer; they are **rhythmic, repetitive, and self-directed**.

A fourth convergence point, drawn from the comparative literature, deserves mention. Terrace (1985) reports that Kanzi, a juvenile bonobo, began using lexigrams spontaneously and without formal training *immediately* following his separation from his mother at age two and a half. Unlike all previously studied apes, who required extensive drilling to acquire each symbol, Kanzi mastered over 30 lexigrams within six months through observational learning alone. The timing is significant: the explosive onset of symbolic behavior was triggered not by a training protocol but by **the rupture of the primary attachment bond**. This case — an artificially scaffolded cross-species rearing situation involving a bonobo, a great ape with more extensive allomaternal tendencies than common chimpanzees, followed by maternal separation and rapid spontaneous symbol acquisition — offers a suggestive analogue to the phylogenetic scenario without making bonobos a cooperative-breeding species in the strict sense.

The ontogenetic data do not prove the phylogenetic scenario, but they demonstrate that the cognitive-affective dynamic proposed — vocalization as a tool for managing the absence of an attachment figure — is a robust and independently documented feature of human development.

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8. Comparative Evidence: Vocal Individuation and Vocal Mimesis

The emergence of displaced reference through the name-leash mechanism requires two distinct biological capacities that are rarely found together: **vocal individuation** (the production of individualized signatures directed at specific others) and **vocal mimesis** (the ability to reproduce another's vocalization). Comparative evidence shows that these building blocks exist separately in different lineages. Their co-occurrence in the context of cooperative breeding is the distinctive contribution of the human lineage.

8.1 Vocal Individuation Without Mimesis: Dolphins and Bats

Bottlenose dolphins (*Tursiops truncatus*) develop individualized "signature whistles" and modify their acoustic properties in the presence of their own calves — producing a dolphin analogue of motherese (Sayigh et al., 2023). Greater sac-winged bats (*Saccopteryx bilineata*) similarly produce pup-directed vocalizations that differ in timbre and peak frequency from adult-directed calls (Fernandez & Knörnschild, 2020). However, neither dolphins nor bats possess the sophisticated social mimesis and mirror neuron systems found in primates. Their “names” remain functional beacons rather than symbolic tools that can circulate among third parties.

A convergent realisation of the broken-index operation, however, has been documented at a deep phylogenetic distance. **Antiphonal duetting** in pair-bonded songbirds — African boubou shrikes, *Cossypha* robin-chats, and other tropical duetting passerines — exhibits the diagnostic phenomenon: when one partner is absent or out of contact, the remaining bird does not simply repeat its own portion of the duet; it **reproduces the missing partner’s unique tune**, and the absent partner returns “as if called by name” (Thorpe, 1972). The signal produced in the partner’s absence is the partner’s signature, not the singer’s own — a vocal designation of an absent specific individual addressed to the silent acoustic environment for the purpose of bringing that individual back. The operation is functionally homologous to the broken-name-leash, in a lineage that diverged from mammals over three hundred million years ago, in a species without cooperative breeding, and in a dyadic rather than networked configuration. What CB and alloparenting in the hominin lineage add is not the operation itself but the *generative* substrate — a network across which the operation can conventionalise, a non-fungible referent of compelling affective weight, and the mimetic infrastructure to spread the signal as a public symbol.

8.2 Vocal Mimesis Without Individuation: Chimpanzees

Chimpanzees (*Pan troglodytes*) exhibit powerful **vocal convergence**. Males gradually adjust the acoustic characteristics of their pant-hoots to match those of their frequent associates, creating a group-level acoustic signature (Mitani & Brandt, 1994; Lombrey & Fröhlich, 2025). This convergence operates within fission-fusion social dynamics and helps maintain social bonds when subgroups separate and reunite. But chimpanzees do not produce individualized vocal signatures directed at specific conspecifics, nor do they modify their vocalizations in infant-directed contexts. Their vocal mimesis is a tool of group cohesion, not of individuation.

8.3 The Marmoset Bridge: Cooperative Breeding and Individualized Vocal Labelling

A particularly instructive case comes from common marmosets (*Callithrix jacchus*) — one of the few primates that, like humans, practice cooperative breeding. Takahashi et al. (2017) demonstrated that **contingent vocal feedback** from caregivers — not simply the overall volume of ambient vocalizations — plays a critical role in bootstrapping vocal development in marmoset infants. The caregiver’s responsive, timed vocalization shapes the infant’s transition from immature to adult-type calls. The marmoset data provide a primate proof-of-concept: cooperative breeding plus vocal plasticity yields individualized vocal exchange.

A subsequent finding by Oren et al. (2024) extends the marmoset case from contingent feedback to **vocal labelling of individual conspecifics**. Using machine-learning classifiers on more than 53,000 spontaneous phee calls, the authors demonstrated that marmosets encode the identity of the intended receiver in the acoustic modulation of the call, and that receivers respond more consistently and correctly to calls directed at them than to calls directed at others. Four features of this system are directly relevant to the Name-Leash architecture.

(i) The label is a modulation within the contact call. The phee call is the species-typical long-distance contact call used precisely when conspecifics are out of sight — a configuration that functionally reproduces the broken index (§3.2). The receiver-specific information is carried by amplitude- and frequency-modulation trajectories *within* the phee call rather than by a distinct lexical unit. This is exactly what the hypothesis predicts: the proto-name emerges from within the maternal–infant vocal contact system, not alongside it.

(ii) The labels are arbitrary, not imitative. Oren et al. specifically tested whether marmoset labels could be explained by vocal accommodation — the well-documented mechanism by which many species modify their own calls to resemble a partner's call. If a caller were merely imitating the receiver's characteristic acoustic signature, then the same classifier trained on caller-to-receiver calls should predict receiver-to-caller responses. The prediction failed. The label is **not acoustically derived from the receiver**. This distinguishes marmoset labels from bottlenose dolphin signature whistles, where labelling operates through imitation of the addressee's own signature — an *iconic* relation grounded in acoustic resemblance rather than convention (Janik, 2000; King & Janik, 2013). The marmoset data thus document a passage to *conventional* reference: a label that is neither triggered by the referent's perceptual presence (indexical) nor copied from the referent's own voice (iconic).

Jaakkola (2025) raised the reasonable objection that vocal accommodation could in principle produce the same classification results without an internal representation of the other. Oren and Omer (2025) provided the decisive response: the arbitrariness test rules out accommodation as sufficient explanation, and marmoset hippocampal neurons independently encode the identity of conspecifics cross-modally, responding to both the phee call and the visual image of the individual (Tyree, Metke & Miller, 2022, 2023; see §9.2). The Jaakkola–Oren exchange is, in effect, an empirical re-enactment of the conceptual dispute between indexical and symbolic reference, conducted on a living cooperative breeder.

(iii) Learning is mimetic and third-party mediated. When Oren et al. tested whether callers learn labels from the receiver's own vocalizations during direct interaction, the prediction again failed. Family members converge on similar labels for a given receiver, including genetically unrelated adults paired as mates. The only mechanism consistent with the data is that marmosets learn labels *by listening to how other family members address the individual in question* — that is, by **observing third-party dyads** in the alloparenting network. This is precisely the mimetic-contagion mechanism proposed in §6.5: the convention does not live in the dyad but in the network, and it spreads because the act of producing the name is itself contagious within a sufficiently dense social structure. The marmoset data demonstrate that mimetic convention, transmitted through third-party observation, does not require shared intentionality between caller and receiver as a precondition.

(iv) Labels exhibit a dialectal rather than a fully cross-group structure. Each receiver is addressed by approximately three distinct labels in the Oren et al. dataset — one per family. Within a family there is convergence; between families the labels diverge. This represents an intermediate point on the continuum between a purely individualized signature (dolphin) and a fully shared public name (human). For the Name-Leash this is a prediction: conventionalization begins locally in dense alloparenting networks and expands outward only when those networks become interconnected.

The experimental caveat is important. The Oren et al. subjects were adult pair-mates and family members, not mother–infant dyads. The asymmetry central to the Name-Leash hypothesis — that *the mother names the infant* rather than the reverse — is not directly tested by this paradigm. What the data establish is that the underlying machinery (arbitrary convention, within-contact-call modulation, network-mediated learning) is present in a cooperatively breeding primate. Whether the maternal–infant dyad drives the initial conventionalization in callitrichids remains an open empirical question.

8.4 Locating Marmoset Contact-Call Modulation on the Decoupling Scale

The Oren et al. result is part of a broader picture of the callitrichid contact call as a multidimensional substrate. A consistent body of work has documented voluntary vocal control in callitrichids across multiple paradigms: controlled phee-call production with adjustable inter-call intervals (Roy et al., 2011), modulation of food-call rate independent of food motivation (Liao et al., 2018), suppression of food calls toward live prey (Pomberger et al., 2020), and Lombard-effect adjustments to ambient noise (Pomberger et al., 2018). Burkart and colleagues (2022), reviewing this literature, conclude that "directly measured individual arousal levels cannot fully explain when [callitrichids] switch between calls." In the language of §2.2, **arousal-decoupling** is empirically established for callitrichid contact calls; the channel is no longer a quasi-automatic readout of the caller's internal affective state.

What flows through the channel once the arousal-readout constraint is relaxed has begun to be characterized. The Oren et al. (2024) finding documents one axis: receiver-specific information that requires a representation of the other (the arbitrariness test, §8.3 (ii); cross-modal hippocampal encoding, §9.2). Mircheva et al. (2026) provide a complementary observation at the level of overall calling rate: contact-call volubility is robustly modulated by social context and information-asymmetry conditions and is not reducible to caller arousal, sex, or status. The phee call in callitrichids is, in these published findings, a multidimensional substrate carrying receiver-individual and rate-level information beyond what an arousal-readout model can account for. **Here-decoupling** — the production of a contextually structured signal in the absence of the perceptual presence of its referent — is the architectural step at which extant callitrichid evidence places the contact call.

What the published comparative record does *not* yet demonstrate, on the analysis offered here, is **now-decoupling**: a vocalization native to one behavioral domain deployed in a state native to a different behavioral domain, with the referent held representationally across an interval that survives shifts in the caller's own context. The proto-name in the Name-Leash architecture is structurally an extreme case of this: a vocalization shaped within the maternal vocal–affective system is sustained while the praxis of the foraging mother is in an entirely different domain (gathering, vigilance, sociality). The hominin step the hypothesis identifies — non-fungibility of the referent, cross-domain use of the vocalization, and now-decoupling of the signaling state — is what cooperative breeding combined with the affective non-fungibility of the infant adds beyond the architecture documented in callitrichids. The phee call as documented in callitrichids is, on this reading, a precise model of the substrate on which displaced reference can subsequently develop, but is not itself the locus where displacement in the strict Hockett-Korzybski sense has occurred.

8.5 The Convergence Gap

The following table summarizes the comparative landscape:

Feature	Dolphins/Bats	Chimpanzees	Marmosets	Early *Homo*
Vocal individuation (signatures/CDC)	YES	NO	YES (contingent)	YES (proto-names)
Vocal labelling of conspecifics	YES (imitative)	NO	YES (arbitrary)	YES (arbitrary, public)
Vocal mimesis (matching/convergence)	LIMITED	YES (group-level)	LIMITED (echopraxic)	YES (mirror system)
Cooperative breeding	NO (dolphins); partial (bats)	NO	YES	YES
Arousal-decoupling	YES	LIMITED	YES (Burkart et al., 2022)	YES
Here-decoupling	YES	YES	YES (Oren et al., 2024)	YES
Now-decoupling	NO	NO	NOT YET DOCUMENTED	YES
Displaced reference (strict sense)	NO	NO	NO	YES

The table illustrates the key claim: the hypothesis treats displaced reference in the strict, Hockett-Korzybski sense as emerging only where all conditions co-occur. Dolphins have individuation and labelling but the label remains indexically bound to the receiver's own voice; their mimesis is limited and CB is absent. Chimpanzees have group-level mimesis but lack individualized labelling and CB. Marmosets — the critical intermediate case — possess arbitrary (non-imitative) vocal labelling of specific others *within* a cooperative-breeding ecology, confirming that arousal-decoupling and here-decoupling can be achieved on the contact-call substrate without the full human cognitive architecture. What the marmoset does not demonstrate, on the published evidence, is the further step: now-decoupling. The hypothesis proposes that only in the hominin lineage did all the conditions converge in a way sufficient for that final step to take place.

9. Neural Substrate: Concept Cells and the Left Temporal Pole

The neural evidence reviewed in this section does not prove the proposed evolutionary pathway. It establishes that the required representational operation — invariant representation of a familiar individual across modalities — is biologically realized in extant primates and therefore constitutes a plausibility anchor for the hypothesis.

The Name-Leash hypothesis requires that the hominin brain possess the neural machinery for two operations: maintaining an invariant representation of a specific individual in the absence of perceptual input, and associating a vocal label with that representation. Contemporary neuroscience identifies two systems that jointly fulfill these requirements.

9.1 Concept Cells: Invariant Identity Representation

The first component is the system of **concept cells** — sparse, highly selective neurons in the medial temporal lobe (MTL), including the hippocampus and amygdala, first described by Quian Quiroga and colleagues (2005; see also Quian Quiroga, 2012, 2017). A single concept cell or small neural ensemble fires selectively for a specific known individual — a particular person, not a category — regardless of the sensory modality of the stimulus: a photograph, a voice recording, or a written name all activate the same cell. This invariance is critical: concept cells provide an **ultra-compact, modality-independent identity code** — precisely the kind of representation that a proper name, as a rigid designator, requires.

Identity-selective neurons with similar properties have been documented in macaques (Chang & Tsao, 2017; Freiwald & Tsao, 2010), demonstrating that the primate brain was already equipped with the neural architecture for individualized representation before the hominin divergence. Behavioural correlates of cross-modal individual recognition extend further across the mammalian line: long-tailed macaques classify conspecifics by social-kinship category, treating mother-offspring pairs as equivalent across many distinct dyads (Dasser, 1988); rhesus macaques spontaneously match the voices and faces of familiar conspecifics (Adachi & Hampton, 2011); domestic dogs recall their owner's face on hearing the owner's voice (Adachi, Kuwahata & Fujita, 2007); and horses show analogous cross-modal individual recognition of human handlers (Proops, McComb & Reby, 2009). The direct connection between concept cells and the amygdala ensures that identity representations of attachment figures are **affectively loaded** — the neural correlate of the separation anxiety that drives the name-leash mechanism.

9.2 Cross-Modal Identity Binding in Cooperative Breeders

A question that remained open in the concept-cell literature concerned species distribution. Initial attempts to document cross-modal identity representations in non-human primates — rhesus macaques in particular — had failed, and this was sometimes taken to suggest that modality-independent identity binding might be uniquely human (Sliwa et al., 2016). Tyree, Metke, and Miller (2022, 2023) overturned this picture. Recording from the hippocampus of common marmosets (*Callithrix jacchus*) while presenting faces and phee calls of familiar conspecifics, they identified a population of single neurons that responded selectively to both the face and the voice of specific individuals — the direct analogue of human concept cells. Cross-modal identity representations are also present in the hippocampus of Egyptian fruit bats (Forli & Yartsev, 2023), which diverged from the hominin line approximately 95 million years ago. The neural machinery for cross-modal individuation of conspecifics is an **ancient mammalian capacity**, not a hominin invention.

Three features of the marmoset finding bear directly on the Name-Leash mechanism. First, the cross-modal binding itself is what the broken-index scenario requires at the neural level: when the infant is visually absent during allomaternal transfer, the concept of the infant — formed through multimodal experience — remains accessible, and a vocalization (the proto-name) operates as a partial cue capable of reactivating that representation. The proto-name is thus not the creation of a new neural function but the **vocal handle** for a pre-existing cross-modal identity concept. Second, the marmoset concept cells exhibit lower selectivity than their human homologues. This is not a problem for the hypothesis but an asset: a gradient rather than a discrete threshold relaxes the requirement that early hominins possessed "fully sharpened" concept cells at the moment of proto-name emergence. The system begins as a graded, distributed representation and is progressively sharpened by the very process the Name-Leash describes — the repeated, affectively weighted retrieval of identity representations under conditions of separation. Third, Tyree et al. observed that identity decoding in the marmoset hippocampus operates robustly at the **population level**: removing the most selective concept cells from the decoder does not meaningfully degrade identity classification, and manifold projections of population activity recover individual identity and **social categories** — family membership is spontaneously recovered from unsupervised clustering of population states. This provides a neural correlate of the **genitive contour** (§§10.1, 10.8): the identity of a known individual is represented in the brain not as an isolated node but as *a position within a relational structure*.

Taken together with the Oren et al. (2024) labelling result (§8.3), this constitutes a partial proto-naming substrate in a single living cooperative breeder. The Name-Leash hypothesis can now be stated in neurally grounded form: **the proto-name is a vocal addressing handle for a pre-existing cross-modal identity concept, deployed at moments when the visual component of that concept has been interrupted by separation, and conventionalized mimetically across the alloparenting network through third-party observation.**

9.3 Cross-Modal Identity Dissociation: Clinical Evidence

The cross-modal architecture has an independent clinical signature. **Prosopagnosia** (Damasio et al., 1990) involves a selective deficit in visual identification of familiar individuals while voice-based recognition remains intact. A particularly diagnostic finding from this syndrome is the dissociation between overt and covert recognition:

prosopagnosic patients who are unable to consciously identify familiar faces nonetheless show normal autonomic responses to those faces — measured as differential skin-conductance to familiar versus unfamiliar individuals (Bauer, 1984). The recognition system is not lost; the conscious-overt route to it is severed while the affective-autonomic route remains intact. **Capgras syndrome** (Ellis & Young, 1990; Hirstein & Ramachandran, 1997) is its mirror image: the patient overtly recognises the face but lacks the autonomic-affective response that normally accompanies recognition; Hirstein and Ramachandran's patient DS displayed a *modality-specific* delusion — claiming his parents were imposters when looking at them but never when speaking to them on the telephone. Voice continued to drive normal identification. The three syndromes — overt-without-affective, affective-without-overt, and modality-specific dissociations — dissociate the visual-overt, visual-affective, and auditory tracts of identity, providing strong clinical evidence that the unified individual-identity representation is *constructed by convergence* from distinct upstream channels. The proto-name is the vocal handle on a representation that can be sustained even when its visual handle is lost.

9.4 The Left Temporal Pole: Name Retrieval and Person Identity

The left temporal pole should not be treated as a literal evolutionary "name module," but it is strongly implicated in proper-name retrieval and person-identity processing. Damasio et al. (1996), in a PET study of lexical retrieval, demonstrated that retrieving the names of unique individuals (faces of famous persons) selectively activates the left temporal pole, while retrieving names of common-noun categories (animals, tools) activates posterior temporal regions instead. Semenza (2006), reviewing the broader neuropsychological literature, documents the same dissociation across multiple methodologies: proper-name retrieval depends on partially distinct neural pathways from common-noun retrieval, and is selectively impaired by focal anterior temporal damage. Subsequent neuropsychological evidence shows that proper names are unusually vulnerable: patients with focal damage to the left temporal pole may retain rich semantic knowledge about a person while losing access to the person's name. This dissociation fits the hypothesis's claim that names are not ordinary predicates but specialized links to invariant individual representations.

For present purposes the point is modest: the brain already contains systems for cross-modal identity binding, and the retrieval of proper names appears to depend on partially specialized neural pathways. The proto-name mechanism would have recruited this architecture rather than inventing identity representation from scratch.

9.5 The Theta-Mediated Consolidation Pathway

A plausible neural bridge between rhythmic vocalization and stable identity representation is theta-mediated consolidation. Auditory cortex entrains to the rhythmic structure of speech, with low-frequency (delta–theta) oscillations phase-locked to syllabic and prosodic rate (Poeppel & Assaneo, 2020); the same theta envelope is the medium within which the gamma-frequency bursts that subserve hippocampal consolidation are organised. From auditory cortex, the relevant signal would propagate forward along the ventral stream

(Saur et al., 2008) to anterior temporal regions implicated in proper-name retrieval. The claim here is not that the proto-name can be directly localized to a single pathway, but that repeated rhythmic vocalization offers a plausible physiological route by which an affectively charged identity representation could be stabilized across absence. This remains a testable prediction rather than a demonstrated fact.

9.6 Evolutionary Mismatch and the "Fragility" of Names

A common observation about proper names is their apparent "fragility" — the frequency of tip-of-the-tongue experiences. The evolutionary perspective suggests the opposite. Names of emotionally close individuals — one's child, mother, partner — are almost never forgotten in healthy adults; they are consolidated through powerful neurochemical mechanisms (oxytocin, noradrenaline) and direct limbic connectivity. Forgetting such names occurs only with gross organic damage. The "fragility" of the name system manifests only at its periphery: when modern humans attempt to store hundreds of names of acquaintances, colleagues, and public figures — a task for which the system was never designed.

The ancestral system, shaped by the pressures of cooperative breeding, was optimized for a small number of **affectively significant individuals** — closer to the inner circle of Dunbar's social network (roughly 5–15) than to the hundreds of names modern life demands. The apparent weakness of the system is, in fact, evidence of its original specialization: it was built for the precise task the hypothesis describes. This has a direct consequence for the debate over internal vs. external storage of lexical items: Tallerman (2014) is correct that I-storage of ordinary words depends on externalization, but the names of attachment figures are maintained through a different mechanism — continuous low-level reactivation by the Default Mode Network (Buckner, Andrews-Hanna & Schacter, 2008) and limbic consolidation — which is independent of public use (see §12.10).

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10. The Genitive Contour as Proto-Syntactic Seed

The term **genitive contour** is used here in a cognitive-relational sense, not as a claim that early hominins possessed grammatical genitive case. It refers to the relational frame in which a name-like vocalization is embedded: not a bare label, but an address or invocation of an individual as *someone's* child, *this* child, or the child-to-be-returned. Grammatical genitives in attested languages are treated below as later synchronic windows onto this relational structure, not as direct fossils of a prehistoric case ending.

10.1 The Name as Inherently Relational

This paper uses the term "Merge" not in the technical Chomskyan sense of formal syntactic combination, but in its literal meaning: **attachment**. The proto-name, we suggest, was born as an inherently attachable unit — a vocal gesture that encoded identity together with

inalienable relation. The infant *was* “Mother’s [Name]” before “[Name]” existed as an independent label; the possessive relation was ontologically prior to the nominal category.

This claim has an anthropological anchor. Annette Weiner (1992) described a class of possessions in Oceanic societies that are **inalienable** — objects that cannot be permanently transferred because they remain defined by their relationship to their original owner. The infant in a cooperative breeding context is the paradigm case of inalienable possession: temporarily held by an allomother, but *remaining* the mother’s child. The situation “my child is not with me” generates a paradox: the possessive relation persists despite the absence of physical proximity. This paradox is, in embryonic form, the genitive construction. The proto-name designates the child within a relation — **the-child-of-this-mother** — carrying its relational context as an intrinsic property.

10.2 Prosodic Evidence: The Motherese Contour

This relational character manifests prosodically in motherese. The name is rarely uttered in isolation; it is embedded in contours of possession (“[Name]’s here?”), search (“[Name]?”), or distress (“[Name]!”). The proto-word carries relational valence as its intrinsic property — much as patronymic surnames fossilize genitive morphology (Ivan-ov, Peter-son, O’Brien), where the boundary between noun and possessive case has eroded, leaving only the relational whole.

10.3 Decomposition Through Perspectival Shift

The proto-name is not imagined here as an isolated atomic label. It emerges in a relational field: mother, infant, allomother, transfer, absence, return. Once the same child is handled and vocalized over by different caregivers, the vocal complex is exposed to repeated perspectival variation. One caregiver gives, another receives; one soothes, another searches; the child remains invariant while relational frames vary.

This creates a plausible decompositional pressure. The invariant element associated with the specific infant can separate from the variable pragmatic and possessive frame. The result is not syntax in the modern sense, but a proto-syntactic condition: a formerly holistic utterance begins to acquire reusable internal components because different participants occupy different relational positions around the same non-fungible individual.

The mechanism is therefore not “Merge” as a primitive operation. It is fission first: the splitting of a relational holophrase under repeated perspectival shifts. Combination becomes possible only after such components have been freed.

10.4 The Proto-Word as “Bead with Handles”

Bickerton’s “beads on a string” metaphor imagines protolanguage as a sequence of isolated lexical items awaiting later syntax. The Name-Leash hypothesis proposes a different image, related to but distinct from the holophrastic protolanguage tradition (Wray, 1998, 2002; Arbib, 2010): the proto-word is a **bead with handles**. It is already relational because it arises from a caregiving configuration, something closer to "my-child / this-child-to-be-returned" than to an isolated label "child." Its later syntactic potential comes from these inherited relational openings.

This distinction changes the order of explanation. Syntax does not have to be added from outside to inert words. Rather, a relational holophrase decomposes, and the freed elements retain traces of the relations from which they came. The earliest combinatorial capacity is therefore not the concatenation of independent atoms but the reuse of elements already shaped by transfer, possession, absence, and return.

10.5 Comparative Ontogenetic Evidence: The Possessive Holophrase

Ontogeny offers a limited but suggestive analogue. Early child speech often treats names, kin terms, and possessive relations as fused pragmatic units before full grammar is available. Such forms do not prove the phylogenetic scenario, but they show that relationally loaded holophrases can precede analytic syntax in development. The relevant point is not that the child repeats prehistory, but that human language acquisition readily begins with socially saturated, person-centered, possessive or caregiving frames rather than with abstract predicates.

10.6 Decomposition and Proto-Syntax

The decomposition of §10.3 creates the possibility of recombination. The freed components, each carrying relational valence, can now be reassembled in new configurations — "[Grandmother's]-[child]," "[Allomother's]-[child]" — producing the first combinatorial structure. In terms of generative grammar, the operation of **Merge** (in its minimal sense: combining two elements into a structured unit) arises as a byproduct of the decomposition of the genitive structure. Syntax does not appear as a separate cognitive innovation; it emerges from the internal logic of the relational name.

A further consideration follows from the rhythmic substrate discussed in §6.4 and §9.5, which would already have been in place before segmentation began. Beat gestures and prosodic contour supply a continuous, pre-semantic temporal glue — a way of asserting "this goes with that" without any explicit linker — exactly the scaffolding that decomposition-first proto-syntax requires. Where Bickerton's "beads on a string" presupposes an implicit grammatical thread, the rhythmic channel supplies an actual, observable one: the theta envelope along which the first lexical separations could propagate.

10.7 Recursion Through Kinship: A Later Expansion

Recursive kinship terminology is best treated as a later expansion of the same relational machinery, not as a prerequisite for the proto-name. Once a name-like sign can hold an individual across absence, relations among such individuals can be stabilized and iterated: mother of mother, child of child, and so on. Grandmotherhood may have amplified this process later in human evolution, but the original name-leash mechanism does not require a fully developed kinship algebra. It requires only a recurrent triadic caregiving field in which the same infant is held, transferred, and returned.

10.8 The Genitive as Fossilized Trace: A Synchronic Window

Modern genitive constructions should not be read as direct fossils of prehistoric grammar. They are, however, useful synchronic windows onto a durable cognitive contour: the absent or dependent entity is held through a relation rather than merely classified as an object. Inalienable possession, kinship reference, partitive and genitive-of-negation systems, and softened request forms all preserve versions of this relational logic. The main point is limited: the proto-name is more plausibly a relational holophrase than an isolated noun. Detailed typological and pragmatic discussion is moved to Supplementary Materials S4.

10.9 Summary of the Evidential Architecture

The argument of this section establishes the genitive contour as the proto-syntactic seed of language not by projecting modern grammar onto a prehistoric stage, but by tracing the reverse direction: from a specific cognitive-ecological situation (maternal separation in cooperative breeding) to the structural traces this situation left in the grammars of attested languages. The genitive is not the literal origin of the word; it is a synchronic register in which the originary situation of relational absence remains legible in later grammatical systems. That this trace appears in typologically diverse forms — case marking, word order, clitics, construct state — is not a weakness but a prediction: the cognitive structure is universal; its morphological clothing is not.

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11. Integration Within Broader Anthropogenesis

The core hypothesis concerns the origin of displaced reference, but it also suggests a wider anthropogenetic implication: reference, obligation, and social time may have differentiated from a single vocal act of holding an absent individual in mind.

11.1 From Name to Obligation

The proto-name is not a neutral label. In the proposed scenario it is tied to transfer and return: the infant is entrusted to another caregiver and expected back. The vocalization that holds the infant in absence therefore also carries a minimal social commitment. This does not mean that language begins with explicit morality or contract. It means only that the first displaced sign may already have been embedded in a structure of trust, care, and deferred return.

11.2 From Debt to Time

Once a sign can hold a specific individual across an interval, social time becomes thinkable in a new way. A promise, a debt, a kinship relation, and a memory all require the maintenance of a relation beyond the present stimulus. The name-leash is the minimal case of this operation: it binds departure to anticipated return. Anthropological and philosophical work on debt as a foundational social category — most prominently Graeber's (2011) historical genealogy and Douglas's (2016) philosophical analysis — has emphasised that the obligation to return precedes monetary economy by many thousands of years and is not derivable from market exchange. The hypothesis advanced here adds a deeper layer: the obligation to return precedes obligation in the social-economic sense and is rooted in the maternal commitment to recover the infant. The broader philosophical reading of sign-as-promise and the human *Umwelt* as time-debt is developed in Supplementary Materials S3.

11.3 The Cultural Fossil Record

Later human cultures preserve many practices that treat names as charged links rather than neutral labels: name taboos, ritualized separation and reunion, repetitive prayer, bead-counting, and mourning practices. These do not prove the evolutionary mechanism, but they are consistent with the phenomenology the hypothesis predicts: to utter a name is to engage an absent or transformed individual across a gap. The ethnographic and cultural material is developed in Supplementary Materials S5.

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12. Comparison with Existing Models

The Name-Leash hypothesis differs from existing accounts of language origins in its candidate for the first displaced signal (a proper name rather than a food-call or a gestural icon) and, no less importantly, in how it explains conventionalization: the process by which a private vocalization becomes a shared symbol. Each major theory proposes, explicitly or implicitly, a different conventionalization mechanism. A systematic comparison reveals that the Name-Leash mechanism is both more parsimonious and more empirically grounded than the alternatives.

12.1 Bickerton: Ecological Recruitment

Bickerton (1990, 2009) correctly identified displacement as the central problem but located the selective pressure in ecological recruitment: a hominin discovers a distant carcass and must communicate its location. The displaced signal, on this account, is a vocal icon. Two problems have already been addressed (§§2–3): the availability of perceptually grounded alternatives (pointing and leading), and the logical circularity of the iconic path. A third problem concerns conventionalization: food is **fungible** — a missed signal costs a meal, not a life — so there is no intense, recurrent pressure on any particular sound to be accurately reproduced. The recruitment model lacks a robust conventionalization engine. The Bickertonian "beads on a string" model has been critically examined in §10.4: pidgin speakers are syntactically competent adults who deliberately simplify; their output is not a model of genuine pre-syntactic communication. The Name-Leash proposes instead that the proto-word was born relational — a "bead with handles" — making combinatorial structure an almost inevitable consequence of decomposition rather than an independent innovation. The recruitment scenario also faces an intergenerational transmission problem: it separates the site of innovation (adult foraging) from the site of acquisition (infant development), creating a transmission gap that the Kawai (1965) two-stage model does not bridge for adult-only signals.

12.2 Terrace: Naming as the Rubicon

Terrace (1985), in *In the Beginning Was the "Name,"* argued in the aftermath of Project Nim that the first generation of ape-language research had pursued the wrong target: by fixating on syntax, researchers had overlooked **referential naming** — the use of a symbol to direct another's attention to an object simply for the purpose of noting it. His title prefigures the central thesis of the present work; his analysis stands in the same relation to the Name-Leash as a diagnosis stands to a treatment.

Four points of contact deserve emphasis. (i) Terrace asks how symbol use could ever become self-rewarding rather than instrumentally motivated; the Name-Leash answers that the first referential act was self-rewarding because it was *self-regulatory* — the reward was endogenous, mediated by respiratory-vagal coupling (§§4, 6.1). (ii) Terrace, like virtually all researchers in the ape-language tradition, frames naming in terms of common nouns; the Name-Leash proposes that the first displaced referential act was a *proper name* for a specific, non-fungible infant, where attachment ensures that the representation is sustained without external reward. (iii) A passage in Terrace (1985) that has received insufficient subsequent attention notes that Kanzi (Savage-Rumbaugh et al., 1985) began using lexigrams spontaneously and without formal training *immediately following his separation from his mother* at two and a half years of age, mastering over 30 lexigrams within six months through observational learning. From the perspective of the Name-Leash, this is a near-experimental confirmation: the rupture of the attachment bond created the affective pressure that drove symbolic behavior. That the subject was a bonobo — a great ape with more extensive allomaternal *tendencies* than common chimpanzees, though not a cooperative breeder in the strict sense (§7.4) — is consistent with a link between alloparenting potential and displaced reference. (iv) Terrace notes (citing Bard & Vauclair, 1984) that ape mothers do not appear to involve their offspring in joint-attention games but does not explain why; the Name-Leash offers a structural explanation: in solitary-breeding great apes, the mother-infant dyad is not embedded in a network of allomaternal care, and there is no infant-mediated selection for vocal fidelity. Terrace describes the symptom; the hypothesis identifies the cause.

12.3 Falk: Motherese as Audio Leash

Falk (2004, 2009) proposed that motherese originated as an "audio leash" to maintain contact between a mother and a temporarily set-down infant. The Name-Leash hypothesis preserves the mother-infant vocal interaction as the crucible of language origins but shifts the critical moment from **contact** (Falk's leash is intact) to **separation** (the leash is broken). It is the rupture of the audio leash, not its maintenance, that generates the pressure for displacement. The hypothesis does not require routine putting-down of infants on the ground during foraging; it relies instead on the well-documented practice of infant transfer to allomothers. Regarding conventionalization, Falk's model faces a specific difficulty: if motherese is a contact-maintaining vocalization between mother and infant, there is no structural reason for it to spread to third parties, since the signal's function is completed within the dyad. The Name-Leash places the infant within a *network* of caregivers, each of whom has independent affective motivation to acquire and reproduce the proto-name — generating conventionalization as an automatic consequence of network structure. Falk herself recognized the relevance of Kawai's (1965) two-stage model of cultural propagation for hominin vocalizations, but her dyadic model does not supply the engine that drives the signal beyond the mother-infant pair into the horizontal network where Stage 1 propagation occurs. The Name-Leash hypothesis supplies the missing engine: regular transfer of the infant to multiple caregivers, and the resulting separation experienced by the mother, force the signal out of the private dyad and into the alloparenting network — triggering both horizontal propagation (among allomothers, Stage 1) and subsequent vertical transmission (through the next generation of infants, Stage 2).

12.4 Tomasello: Shared Intentionality

Tomasello (2008) emphasizes shared intentionality and joint attention as the cognitive foundation of language: pointing gestures → shared attention → pantomime → gradual vocalization → convention. The model faces two difficulties when applied to the *origin* of displaced reference. First, it describes a synchronic cooperative stance — two agents attending to the same object in the same moment — and does not, by itself, explain how reference could operate in the absence of the referent. Second, it requires the capacity for mutual perspective-taking as a *precondition* for conventionalization, making the explanatory order circular. The Name-Leash hypothesis changes this order (§6.7): shared intentionality need not be the foundation of conventionalization and may instead be its **emergent product**. When multiple alloparents independently acquire the same proto-name through mimetic contagion, and each uses it in interaction with the same infant, a convergence of perspectives on a single referent has been achieved without any party needing to understand or represent the other's perspective. The infant becomes the invariant at the intersection of multiple viewpoints — the object is constituted through the convergence, not presupposed by it. Tomasello's shared intentionality is, on this account, a retrospective description of a state already achieved through sub-intentional mechanisms.

12.5 Deacon: The Symbolic Species

Deacon (1997) offered a sophisticated account of symbolic reference in Peircean terms, arguing that symbols emerge through networks of cross-referencing indexical associations, and located the origin of symbolic communication in the marking of social contracts — pair-bonding agreements among males and females in early hominin groups. Deacon's account stands in continuity with Harnad's (1996) symbol-grounding programme, which posed the question of how arbitrary symbols can acquire meaning at all without being parasitic on already-meaningful representations. Deacon's semiotic architecture is powerful, but his conventionalization mechanism requires a top-down process of group-level enforcement: the entire community must simultaneously recognize and sanction the symbolic contract. This is sociologically implausible at the earliest stages — it presupposes the very social-contractual machinery that the symbols are supposed to create. The Name-Leash hypothesis can be read as supplying the missing scenario for both Harnad's grounding problem and Deacon's symbolic threshold. The broken index, repaired by the proper name, is the concrete mechanism through which the symbol acquires its grounding: the proto-name is grounded in the affective-perceptual history of the specific infant it designates, not parasitically on a prior symbol system. Conventionalization is *bottom-up*: it arises from the automatic operation of mimetic contagion in a care network, reinforced by the infant's differential calming response, rather than from any group-level deliberation. The infant is a more *tangible* first object of symbolic reference than an abstract pair-bonding contract.

12.6 Porshnev: Interdiction

Porshnev (1974) proposed that the first conventional signal was an **interdiction** — a vocal command that blocks another's behavior by triggering, through imitation, the inversion of an inhibitory dominant. The Name-Leash hypothesis shares with Porshnev three fundamental insights: the neurophysiological substrate of displacement activities as the reservoir of material from which signals are drawn; mimetic contagion as the mechanism of transmission; and the phylogenetic intersection of an ascending curve of displacement behaviors with an ascending curve of imitative power as the critical evolutionary innovation. The divergence is in the **ecological context**. Porshnev locates the critical mechanism in agonistic interaction (dominance, control); the Name-Leash relocates it to **cooperative care**. The first signal is not a command but a mantra; its function is not to block another's behavior but to self-regulate the vocalizer's affect; its spread through the group is driven not by antagonism but by the universal mammalian response to infant appeal. This relocation has several advantages. Evolutionary biology generally finds it easier to explain complex communicative systems from cooperation than from antagonism. The infant provides a stronger conventionalization engine than the dominant-subordinate relationship, because the infant's differential calming response creates a positive reinforcement loop absent in the interdiction scenario. And the relational (genitive) structure of the proto-name, which provides the seed of proto-syntax, has no analogue in the interdiction model. Porshnev's contribution remains indispensable; the hypothesis can be read as a recontextualization of Porshnev's neurophysiology within the ecological framework of cooperative breeding.

12.7 Saussure: The Arbitrariness Axiom

The models above are all *origin scenarios*. Saussure's contribution is of a different kind: a theory of the sign whose foundational assumption — the arbitrariness of the bond between signifier and signified — has shaped the entire field's understanding of what a sign is.

Zimmerling (2025), reconstructing the implicit logic of Saussure's generalization, shows that it follows an ascending ladder of abstraction: personal names → all proper names → all names → all content words → all linguistic signs. The principle of arbitrariness is, in effect, an *extrapolation* of the observation that personal names are conventional labels — extended step by step to the entire system. The proper name is thus not a marginal anomaly within a system whose norm is the common noun (as Kuryłowicz [1962] and Chomsky [1965] suggested) but the *paradigm case* from which the general principle was secretly derived. The Name-Leash hypothesis reads the ladder genetically rather than analytically: the proper name is first not in the order of description but in the order of emergence, and its "arbitrariness" is the residue of a causal history — the breaking of the indexical bond, the detachment of the vocalization from the perceptual situation, and the overwriting of the original affective motivation through mimetic conventionalization. The sign *appears* arbitrary because its history has been erased by convention, just as a coin whose relief has been worn smooth by circulation appears to be an inherently featureless disk. Arbitrariness is what the sign looks like when its diachronic depth is collapsed into a synchronic snapshot. The broader cultural-semiotic implications of this reading are developed in Supplementary Materials S1.

12.8 Comparative Summary: Conventionalization Mechanisms

Model	First signal	Conventionalization mechanism	Direction	Reinforcement	Infant-mediated transmission
Bickerton	Vocal icon (food)	Unspecified (utility?)	—	Weak (food is fungible)	No: adult foraging context
Terrace	Name (common noun)	Innate primitive + convention	—	Absent (recognized as unsolved)	No: laboratory paradigm
Falk	Motherese	Dyadic vocal exchange	Bottom-up	Moderate (contact maintenance)	Partial: confined to dyad
Tomasello	Pointing gesture	Shared intentionality	Top-down	Cognitive (mutual understanding)	No: presupposes adult capacities
Deacon	Ritual symbol	Group-level enforcement	Top-down	Social (collective sanction)	No: ritual context is adult
Porshnev	Interdiction	Mimetic contagion	Bottom-up	Negative (behavioral blocking)	No: agonistic context
Wray	Holistic command	Fractionation (phonetic overlap)	Bottom-up	Weak (communicative utility)	No: adult command context
Saussure	(not specified)	Axiomatic ("social fact")	—	Structural	N/A: no origin proposed
Name-Leash	Proper name (mantra)	Mimetic contagion in network	Bottom-up	Positive (infant calming)	**Yes**: infant is arbiter and bridge

The Name-Leash mechanism is unique in combining bottom-up automaticity (shared with Porshnev, partially with Falk) with positive reinforcement (the infant's differential response), network amplification (the alloparenting "market"), and inherent relational structure (the genitive contour that seeds proto-syntax). No other model provides all four.

12.9 Wray: Holistic Protolanguage and the Displaced Reference Crisis

Alison Wray's holistic protolanguage model (Wray, 1998, 2000, 2002) constitutes the most developed theoretical framework for understanding the pre-linguistic communicative system from which language emerged, and provides the most direct point of comparison with the Name-Leash hypothesis.

Points of convergence. Three central conclusions are shared. First, both models identify the **demand for displaced reference** as the critical pressure that breaks the pre-linguistic system: Wray's holistic protolanguage works perfectly in the here-and-now but fails when the referent is absent; the Name-Leash identifies the daily maternal-infant separation in cooperative breeding as the ecological context that makes this failure chronic and inescapable. Second, both identify the **proper name** as the first discrete referential unit — Wray through logical reconstruction (the "back-door" proper name that emerges from individualized commands such as *baku*, "bring-Mary"), the Name-Leash through biological specification (the mother's affective vocalization for her non-fungible infant). Third, both predict that the appearance of the name triggers a cascade producing proto-syntax: in Wray's scenario, juxtaposition of a name (*baku*) with a holistic command (*tebima*) transforms the command into a proto-verb with argument structure; in the Name-Leash, decomposition of the genitive holophrase releases relational components that recombine into the first syntactic structures.

Points of divergence. The models differ in three respects that are complementary rather than contradictory. (i) *Motivation.* Wray describes the logical structure of the displaced-reference crisis but does not specify what ecological pressure made it chronic. The Name-Leash fills this slot: cooperative breeding produces daily, predictable, hormonally driven separations from a non-fungible infant — the most powerful possible affective engine. (ii) *Mechanism of conventionalization.* Wray's fractionation model relies on accidental phonetic overlap between unrelated holophrases — a process that, as Tallerman (2007) incisively argued, presupposes the analytic capacity it is meant to explain. The Name-Leash proposes an alternative: perspectival variation in the alloparenting network, where the same sound accompanies different actions (giving, taking, holding, returning), and the invariant (the infant's identity) separates from the variable (the relational frame) through domain-general statistical extraction (Saffran, Aslin, & Newport, 1996). This avoids Tallerman's circularity objection because it operates on pragmatic context, not phonetic substrings. (iii) *Neurocognitive substrate.* Wray's model remains at the level of behavioral logic. The Name-Leash grounds the proto-name in concept cells and vagal regulation, and locates it on the dual-process architecture that Wray herself helped describe (Wray, 2002, 2008; Van Lancker Sidtis, 2004, 2021) — supplying the neural and evolutionary grounding she left open, rather than disputing that account.

Wray's self-imposed paradox. In her 1998 paper, Wray observed that the presence of personal names in a communication system would "automatically" entail grammar, and concluded that early protolanguage therefore lacked names. The present hypothesis accepts the conditional and draws the opposite inference: the name does not presuppose grammar; it generates it. What Wray identified as a reductio ad absurdum is, from the decomposition-first perspective, a description of the syntactogenic power of the proper name.

Relationship to Tallerman's critique. Tallerman (2007) raised three objections to holistic protolanguage: the fractionation paradox; the phonological instability of long unstructured sequences; and the "neurological gap" between subcortical primate vocalizations and cortical human speech. The Name-Leash addresses all three. The fractionation paradox is resolved by replacing phonetic analysis with pragmatic invariant extraction (§10.3). Phonological instability is mitigated by the formulaic character of the proto-name: as a holistic chunk processed by the right hemisphere and basal ganglia (Van Lancker Sidtis, 2004), it is inherently resistant to articulatory drift. The neurological gap is bridged by the dual-process model itself: the proto-name begins as a subcortical-right-hemispheric formulaic unit and is later co-opted by left-hemispheric cortical systems as analytic grammar develops — a migration, not a leap.

Wray's holistic protolanguage and the Name-Leash hypothesis are not competing models but descriptions of successive stages and complementary levels of the same process. Wray describes the communicative system that *precedes* and *surrounds* the emergence of the first name; the Name-Leash describes the biological mechanism by which the first name *emerges from* that system. The convergence of the two — arriving at the same semiotic object (the proper name) from different starting points — constitutes mutual confirmation.

12.10 Tallerman (2014): The Lexicon, Edge Features, and Two Kinds of Displacement

Tallerman's critique of holistic protolanguage is important because she argues that fractionation presupposes the analytic capacities it is meant to explain. Her central target is the Strong Minimalist Thesis (Berwick & Chomsky, 2011), which she contests on the grounds that I-storage of lexical items, far from being autonomous, depends on externalisation: studies of language attrition show vocabulary decaying rapidly without public use (Schmid, 2011). The Name-Leash hypothesis avoids the circularity she identifies by making decomposition pragmatic before it is phonological: the invariant infant is extracted from variable caregiving frames in the alloparenting network. Edge features are not assumed at the start; they accrue as a relational holophrase is reused in different contexts.

12.11 Hurford and the R/C-Predication Threshold

Hurford's (2001, 2003) argument that protothought lacked logical names marks the opposite pole from the present proposal. The Name-Leash hypothesis agrees with Hurford that *predication* is a later achievement, but denies that *individual reference* must therefore be late. A proper name can be referential without being predicative: it can hold a specific individual before the system has subjects, predicates, or propositional truth conditions. The point is reinforced by two BBS commentaries on Hurford's target article. Bickerton (2003) observed that animals "have a clear concept of a specific individual" — the cognitive infrastructure for individual reference is broadly distributed and need not be derived from logical predicates. Dessalles and Ghadakpour (2003) drew the orthogonal distinction between R-predication (recognition / categorisation) and C-predication (contrast operating through explicit negation), and argued that C-predication is the distinctively human achievement; the maternal mantra, on the present account, is the primitive C-predicative act ("not just any infant — *mine*, now *not here*"). Carstairs-McCarthy (2003) further questioned whether Hurford's predicate-and-variable architecture must be invoked at all to explain what is observed in animal cognition.

12.12 Jenkins (2013) and the Tartu Lineage

Jenkins's (2013) discussion of names, protosymbols, and the Tartu-Moscow tradition supports a different ordering of the standard account: proper names need not be treated as marginal or defective nouns. They may instead preserve the trace of a pre-grammatical sign type. Jenkins's typology draws on Morris's (1971) general theory of signs and on Sebeok's (1986) zoosemiotic concept of *identifiors* — signs that communicate information about the nature of their bearers, of which the singular personal name is the limit case. The same line connects to Wilson's (1980) distinction between "personal" and "impersonal" social systems, in which individual recognition is the trait from which all advanced vertebrate sociality is said to flow. The Name-Leash hypothesis stands in this lineage: it accepts the Tartu-Moscow diagnosis that the proper name belongs to a stratum prior to the descriptive lexicon, accepts the zoosemiotic extension of identifiors across the animal kingdom, and proposes the developmental-affective mechanism by which the specifically human form of the displaced identifior emerged. The fuller cultural-semiotic discussion belongs in Supplementary Materials S1.8.

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Limitations

The hypothesis presented here is conceptual. It proposes a plausible evolutionary mechanism connecting cooperative breeding, maternal separation anxiety, and the emergence of displaced symbolic reference, but it does not claim direct empirical demonstration. Several limitations should be acknowledged.

First, the causal chain is inferred from the convergence of evidence across disciplines — primatology, developmental psychology, neuroscience, semiotics, ethnography — rather than from any single line of direct proof. No archaeological or fossil evidence can directly attest to the vocalizations of Pleistocene mothers, and the affective-physiological substrate proposed in §§4 and 6 cannot be reconstructed from the skeletal record. The argument's strength rests on the mutual constraint imposed by independent evidential lines, not on any one of them.

Second, the hypothesis depends on the chronological priority of cooperative breeding in the hominin lineage. The "CB-first" model (Burkart et al., 2025) and the broader case marshalled by Hrdy (2009) provide strong support, but the timing — and the question of whether human CB arose by a facilitating versus forcing driver (Supplementary Materials S2) — remains contested. If a more recent timing for human CB were established, the temporal window for the proposed mechanism would compress correspondingly.

Third, the developmental parallels adduced in §7 (babbling, Fort/Da, Porshnev's first word) are suggestive but do not constitute proof of phylogenetic recapitulation. Ontogenetic patterns may reflect domain-general cognitive dynamics rather than a recapitulation of evolutionary sequence. The convergence with Kanzi's case (§§7, 12.2) is striking but rests on a single individual.

Fourth, the semiotic argument (§3) is a logical reconstruction, not an empirical observation. It identifies a *possible* pathway from index to symbol — the broken-index pathway — not the only one. Alternative pathways (iconic, gestural, ritual-symbolic) remain logically available even if, as argued in §§3.1 and 12, they face independent difficulties.

Fifth, the predictions of the neural model (§9.5) are testable in principle but await targeted neuroimaging, behavioural, and ethnographic work. The hypothesis stands or falls in part on the eventual outcome of these tests.

Despite these limitations, the hypothesis is constructed so that its central claims can be evaluated empirically rather than merely contemplated.

Competing Interests

The author declares no competing interests.

Data Availability

This paper presents a theoretical synthesis. No primary empirical data were generated. All sources are cited in the references.

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